

A Project Report
deessa Companion
Nepal's First Offline Nepali Autism Support App

In partial fulfillment for the degree of BCA under
Pokhara University

Submitted to
Oxford College of Engineering and Management
Bachelor of Computer Application (BCA) Program

Under the guidance of
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ROLES AND RESPONSIBILITIES

This section outlines the specific roles and responsibilities of each team member involved in the deessa Companion project. Clearly defined roles and responsibilities supported effective collaboration, accountability, and smooth project execution. Each member contributed to different aspects of application development, design, testing, and documentation.

Team Member	Role	Responsibilities
Farhan Alam	Project Lead / FullStack Developer	• Coordinating project activities and development tasks. • Developing backend services, APIs, and database integration. • Managing system architecture and core technical implementation.
Rejina Gharti Magar	Frontend Developer	• Implementing application interfaces and frontend functionality. • Integrating frontend components with backend APIs and services. • Developing and refining interactive features for the application.
Binusha Kandel	UI/UX Lead / Android Developer	• Designing user-friendly, accessible, and consistent application interfaces. • Conducting functional testing and reviewing the system for usability and quality.

STUDENT’S DECLARATION

We hereby declare that the project report entitled “deessa Companion: Nepal’s First Offline-Capable, Nepali-Language Autism Support Mobile Application,” submitted in partial fulfillment of the requirements for the degree of Bachelor of Computer Application (BCA) under Pokhara University, is our original work carried out under the guidance and supervision of the designated project supervisors. To the best of our knowledge, this report contains no material that has been previously published or written by any other person, except where due acknowledgment has been made through appropriate citations and references. All sources of information, ideas, and materials used in the preparation of this report have been duly acknowledged and listed in the References section.

We further declare that this project report, in whole or in part, has not been submitted previously, nor is it being submitted concurrently, for the award of any degree, diploma, or other academic or professional qualification at Pokhara University or any other institution.

We accept full responsibility for the authenticity, accuracy, and originality of the work presented in this report.

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SUPERVISOR'S RECOMMENDATION

We hereby recommend that the project report entitled “deessa Companion: Nepal’s First Offline-Capable, Nepali-Language Autism Support Mobile Application,” prepared and submitted by Farhan Alam, Rejina Gharti Magar, and Binusha Kandel, students of the 8th Semester of the Bachelor of Computer Application (BCA) program under Pokhara University, is an original work carried out under our guidance and supervision.

The project has been completed in partial fulfillment of the requirements for the degree of Bachelor of Computer Application (BCA). Throughout the development of this project, the students have demonstrated their ability to identify a relevant problem, analyze its requirements, design and implement a software solution, and conduct appropriate testing and documentation.

To the best of our knowledge, the work presented in this report is satisfactory in scope and quality and fulfills the academic requirements prescribed by Pokhara University for the BCA project. We therefore recommend this project report for final evaluation and examination.

Anil Thapaliya

Project Supervisor

Date: 10th September 2026

Shiva Bahadur Pathak

Project Supervisor

Date: 10th September 2026

LETTER OF APPROVAL

We hereby certify that we have examined the project report entitled “deessa Companion: Nepal’s First Offline-Capable, Nepali-Language Autism Support Mobile Application,” prepared and submitted by Farhan Alam, Rejina Gharti Magar, and Binusha Kandel in partial fulfillment of the requirements for the degree of Bachelor of Computer Application (BCA) under Pokhara University.

We have also examined the project work and evaluated the students’ project defense. Based on the examination and evaluation, we are satisfied that the project has been completed in accordance with the academic requirements of the BCA program and is satisfactory in terms of its scope, quality, and implementation.

We therefore approve this project work as fulfilling the requirements for the Bachelor of Computer Application (BCA) degree under Pokhara University.

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Date:	Date:	Date:
10 th September , 2026	10 th September, 2026	10 th September, 2026

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We would like to express our sincere gratitude to everyone whose guidance, support, and encouragement contributed to the successful completion of this project.

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Finally, we express our deepest gratitude to our parents and families for their unwavering love, patience, encouragement, and support throughout our academic journey. Their constant motivation gave us the strength and determination to overcome challenges and bring this project to completion. To everyone who contributed directly or indirectly to the completion of this project, we extend our heartfelt thanks and appreciation.

ABSTRACT

deessa Companion is an offline-capable, Nepali-language digital platform developed to support children with Autism Spectrum Disorder (ASD), their families, and caregivers in Nepal. It addresses the limited availability of affordable, accessible, and culturally appropriate autism support technologies for communication, learning, emotional well-being, and progress monitoring. The system was developed using Agile methodology and follows a three-tier architecture comprising a native Android application using Java and XML, a Django REST Framework backend, and a Next.js administration panel. An offline-first approach using Room for local SQLite storage and WorkManager for synchronization enables core features during network interruptions. The application includes authentication, picture-based Augmentative and Alternative Communication (AAC), interactive learning, feelings and emotions, progress monitoring, rewards, and routine notifications. The AAC module provides 30 bilingual picture cards with audio support, while learning activities cover alphabets, sounds, shapes, numbers, and colors. The feelings module includes emotion check-ins, breathing exercises, meditation, calming music, and coloring activities.

Testing demonstrated that the system met its functional and performance objectives, with 382 backend test cases and a complete pass rate for implemented Android unit tests. Overall, deessa Companion demonstrates the potential of an offline-first, multilingual platform to improve access to autism support resources and provide a practical foundation for technology-assisted care and learning in Nepal.

Keywords: Autism Spectrum Disorder, Augmentative and Alternative Communication (AAC), Nepali Language, Offline-First, Mobile Application, Android, Django, Accessibility, Child Development.

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AAC	Augmentative and Alternative Communication
API	Application Programming Interface
APK	Android Package Kit
ART	Android Runtime
ASD	Autism Spectrum Disorder
BCA	Bachelor of Computer Applications
CRUD	Create, Read, Update, Delete
CORS	Cross-Origin Resource Sharing
DB	Database
DRF	Django REST Framework
ERD	Entity Relationship Diagram
FCM	Firebase Cloud Messaging
FK	Foreign Key
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IDE	Integrated Development Environment
IEP	Individualized Education Program
JSON	JavaScript Object Notation
JWT	JSON Web Token
MIME	Multipurpose Internet Mail Extensions
MVVM	Model-View-ViewModel

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CHAPTER 1 : INTRODUCTION

1.1 Background

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that can affect communication, social interaction, behavior, and learning. According to the World Health Organization (WHO), approximately one in 100 children worldwide is affected by autism, highlighting the need for accessible and continuous support for children with autism and their families.

In Nepal, families of children with autism may face challenges related to access to specialized services, affordability, language, geographical location, and availability of suitable learning and communication resources. Professional services are concentrated mainly in urban areas, creating additional difficulties for families in other regions. Technology can provide support through Augmentative and Alternative Communication (AAC), structured learning activities, emotional-support tools, and progress monitoring. However, many existing solutions are designed for international users and may not adequately address the linguistic, cultural, financial, and connectivity conditions of Nepali families.

Reliable internet connectivity is also not uniform across Nepal. Therefore, applications that depend entirely on continuous internet access may provide limited support in areas with unstable connectivity. An offline-capable approach can help maintain access to essential communication and learning features when internet connectivity is unavailable.

To address these needs, deessa Companion was developed in collaboration with the deessa Foundation as a mobile-based autism support and learning platform for the Nepali context. The application follows an offline-first approach and focuses on communication, learning, and progress tracking, with additional features for feelings and emotions, rewards, routines, notifications, and caregiver support. It is intended to complement professional therapy, medical care, and educational intervention by providing accessible tools for children and their caregivers in everyday environments.

1.2 Objectives

The primary objectives of the project are:

- **To develop a Nepali-language autism support application** that provides accessible communication, learning, and progress-tracking tools for children with autism and their caregivers.
- **To implement an offline-first architecture** that allows essential application features to remain available during periods of limited or unavailable internet connectivity.
- **To provide picture-based AAC functionality** that enables children with communication difficulties to express needs, emotions, activities, and other common requirements using visual cards with Nepali and English audio support.
- **To develop interactive learning activities** covering alphabet recognition, color matching, number recognition, shape recognition, and animal sounds through structured and gamified activities.
- **To provide progress tracking and reporting** through statistics, dashboards, charts, and reports that allow parents and caregivers to monitor participation and development.
- **To implement a reward and achievement system** using stars, badges, milestones, and related gamification elements to encourage children's participation in learning activities.
- **To develop a web-based administrative panel** for managing application content, users, rewards, and analytics.

1.3 Purpose, Scope, and Applicability

1.3.1 Purpose

The purpose of deessa Companion is to provide an accessible and user-friendly digital support platform for children with Autism Spectrum Disorder and their parents or caregivers in Nepal. The application is designed for use in everyday and home-based environments to support communication, learning, emotional activities, and progress monitoring.

The system focuses on three primary areas:

- **Communication:** Picture-based AAC cards help children express basic needs, emotions, activities, and common requirements.
- **Learning:** Structured interactive activities support basic educational and cognitive skills.
- **Progress Tracking:** Activity and communication information is recorded and presented through visual dashboards and reports.

Supporting features include feelings and emotions, routines, schedules, medications, notifications, rewards, achievements, and emergency contacts. The application supports Nepali and English and is designed to retain access to important functionality during periods without internet connectivity.

The application is intended to complement rather than replace professional therapy, medical care, or educational intervention. Its purpose is to extend opportunities for communication, learning, and engagement beyond formal support environments.

1.3.2 Scope

The project covers the design, development, integration, and testing of three major components:

Table 1.1 Table of Scope

Component	Main Technologies	Primary Purpose
Android Mobile Application	Java, XML, Android Studio	Communication, learning, progress, child and caregiver support
REST API Backend	Django, Django REST Framework	Authentication, data management, synchronization, and application services
Web Admin Panel	Next.js, React, TypeScript, Tailwind CSS	Content, user, reward, and system administration

1.3.2.1 Android Mobile Application

The Android application is primarily designed for children with autism and their parents or caregivers. It provides separate parent-oriented and child-oriented experiences.

The parent interface provides five main navigation areas:

- **Home:** Dashboard, child selection, schedules, notifications, and care-related information.
- **Talk:** Picture-based AAC cards with categories, bilingual labels, audio playback, search, filtering, favorites, and usage information.
- **Learn:** Interactive activities covering Color Match, Shape Explorer, Count & Learn, Alphabet Fun, and Animal Sounds.
- **Progress:** Activity statistics, progress information, trends, reports, milestones, and achievements.
- **Profile:** Child and caregiver management, schedules, medications, emergency contacts, account settings, language preferences, and notifications.

The application also provides a protected Child Mode with a simplified interface focused on Talk, Learn, Feelings, and Rewards. A parent-controlled PIN separates child-oriented activities from parent management functions.

1.3.2.2 Backend REST API

The backend is implemented using Django and Django REST Framework (DRF). It provides server-side services for authentication, child and parent profiles, AAC content, learning activities, emotions, routines, medications, rewards, progress, notifications, synchronization, audit logging, and analytics. SQLite is used during development, while PostgreSQL is intended for production deployment.

1.3.2.3 Web Admin Panel

The web-based administrative panel is developed using Next.js, React, TypeScript, and Tailwind CSS. It allows authorized administrators to manage application content and system data, including AAC cards, learning activities, rewards, users, child-related information, and analytics.

1.3.3 Applicability

The system is applicable in the following contexts:

- **Home-based support:** Parents and caregivers can use the application to support communication, learning, emotions, routines, and daily engagement.
- **Parent-assisted learning:** Caregivers can support structured learning activities and monitor participation at home.
- **Child-centered activities:** Children can independently or with assistance use AAC, learning, feelings, and reward features according to their abilities.
- **Limited-connectivity environments:** Offline-capable functionality helps maintain access to important features when internet connectivity is unavailable.
- **Caregiver monitoring:** Parents can manage child information, schedules, medications, emergency contacts, and progress from a centralized interface.
- **Administrative management:** Authorized administrators can manage content, users, rewards, and system information through the web panel.

1.4 Achievements

The project has achieved the following major milestones:

1. **Developed an integrated three-component platform** consisting of a native Android application, Django REST API backend, and web-based administrative panel.
2. **Implemented parent and child experiences** with a five-section parent interface and a simplified PIN-protected Child Mode.
3. **Implemented a bilingual AAC communication system** using categorized picture-based cards, Nepali/English labels, audio support, search, filtering, favorites, recent cards, and usage tracking.
4. **Implemented five interactive learning activities**—Color Match, Shape Explorer, Count & Learn, Alphabet Fun, and Animal Sounds—with structured activities, difficulty levels, scoring, and rewards.
5. **Implemented progress, emotional-support, and reward features**, including activity statistics, reports, emotion check-ins, calming activities, stars, badges, milestones, and achievement tracking.

1.5 Organization of Report

The remainder of this report is organized into seven chapters.

Chapter 1: Introduction presents the background, objectives, purpose, scope, applicability, major achievements, and organization of the project.

Chapter 2: Survey and Reviews presents relevant literature and technology reviews related to autism-support technologies, AAC, mobile applications, offline-first systems, accessibility, and gamification. It also reviews existing systems and identifies relevant gaps.

Chapter 3: Requirements Analysis describes the problem addressed by the project and presents the functional and non-functional requirements, project planning, resource requirements, product requirements, and conceptual models.

Chapter 4: System Design presents the architecture and design of the proposed system, including the mobile application, backend, API, database, navigation structure, and user interface.

Chapter 5: Implementation and Testing describes the implementation of the Android application, backend, and administrative panel. It also presents key implementation techniques and the overall testing approach.

Chapter 6: Results and Discussion presents the results of system testing and evaluation, including performance, usability, application interfaces, and relevant implementation outcomes.

Chapter 7: Conclusion summarizes the project outcomes and discusses its significance, limitations, and future scope, including potential extensions for therapist and teacher functionality.

CHAPTER 2 : SURVEY AND REVIEWS

2.1 Literature Survey

Technology has increasingly been used to support individuals with Autism Spectrum Disorder (ASD) in areas such as communication, learning, engagement, and progress monitoring. Previous research has explored Augmentative and Alternative Communication (AAC), mobile applications, offline-capable systems, and gamification. These studies provide the foundation for the design of deessa Companion and highlight the importance of accessibility, localization, usability, and reliable access.

2.1.1 Augmentative and Alternative Communication (AAC)

Light and McNaughton (2012) emphasized the role of Augmentative and Alternative Communication (AAC) in supporting individuals with complex communication needs. Beukelman and Light (2020) further discussed the development of AAC from traditional communication boards to digital speech-generating systems, highlighting the importance of simple navigation, personalization, consistent interfaces, and suitable vocabulary.

These findings support the picture-based AAC module in deessa Companion, which uses categorized communication cards, bilingual labels, and audio feedback to provide an accessible communication method for children with speech-related difficulties.

2.1.2 Mobile Applications for Autism Support

Walsh et al. (2017) reviewed mobile applications for autism and identified communication and social-skill support as major application areas. Laski et al. (2018) also reported positive communication-related outcomes from tablet-based picture communication applications and highlighted the importance of visual consistency, audio feedback, and age-appropriate content.

These findings demonstrate the potential of mobile technology to provide portable autism support. deessa Companion applies these principles through a child-friendly interface, Nepali and English content, visual communication cards, audio feedback, and structured learning activities.

2.1.3 Offline-First and Low-Connectivity Solutions

Brewer et al. (2015) emphasized the importance of systems that can continue providing essential functionality during network interruptions. Ratanasanyasak et al. (2020) highlighted local data storage, synchronization, and graceful handling of connectivity loss when designing mobile systems for low-connectivity environments.

These findings are relevant to deessa Companion because its core features are designed to remain accessible without continuous internet connectivity. Local storage and synchronization support continuity of use and data synchronization when connectivity is restored.

2.1.4 Gamification in Education

Deterding et al. (2011) described gamification as the use of game-design elements in non-game contexts, including points, badges, and progress indicators. Hamari et al. (2014) found that gamification can improve engagement and motivation, although its effectiveness depends on appropriate implementation.

deessa Companion uses stars, badges, milestones, and reward streaks as supporting mechanisms to encourage participation in learning activities while keeping learning and skill development as the primary objectives.

2.1.5 Technology for Autism in Developing Countries

Research on autism-support technologies in developing countries identifies language, culture, affordability, accessibility, and connectivity as important design considerations. El Ghazali et al. (2018) emphasized localization and cultural context, while Khan (2019) discussed challenges in autism support in countries including Nepal and highlighted the potential of mobile technology to extend access to support.

These findings support the need for a localized solution for Nepal. deessa Companion addresses these considerations through bilingual content, offline-capable functionality, child-friendly interfaces, AAC communication, structured learning, and parent-oriented progress monitoring.

2.2 Review of Similar/Relevant Projects

2.2.1 Proloquo2Go

Proloquo2Go is an AAC application developed by AssistiveWare for individuals with complex communication needs. It provides symbol-based communication, speech output, vocabulary resources, and customization options.

Strengths:

- Extensive symbol and vocabulary resources.
- High degree of communication customization.

Limitations:

- Does not provide Nepali-language support.
- Primarily focuses on AAC rather than integrated learning, emotional support, and progress monitoring.

2.2.2 Avaz AAC

Avaz AAC is a picture-based AAC application developed in India. It provides speech output, customizable vocabulary, and support for multiple languages, making it relevant to the South Asian context.

Strengths:

- Picture-based communication with speech output.
- Designed with consideration for South Asian users and multiple languages.

Limitations:

- Does not specifically provide Nepali-language support.
- Primarily focuses on communication rather than an integrated learning, emotional-support, and progress platform.

2.2.3 Echo Autism

Echo Autism is a Nepal-focused initiative associated with autism awareness and support services. Its local focus makes it relevant when considering autism-related resources available within Nepal.

Strengths:

- Focuses on the Nepali context.
- Contributes to local autism awareness and support.

Limitations:

- Primarily focuses on support and resources rather than a child-focused mobile platform.
- Does not provide the integrated AAC, learning, emotional-support, and progress features proposed by deessa Companion.

2.3 Comparison Between Existing Systems and deessa Companion

Table 2.1 - Table of Comparison of Existing Systems with deessa Companion

Feature	Proloquo2Go	Avaz AAC	Echo Autism	deessa Companion
Nepali Language	No	No	Context- focused	Yes
Offline Capability	Partial	Partial	Limited	Yes
AAC Communication	Yes	Yes	Limited	Yes
Learning Activities	No	Limited	No	Yes
Progress Tracking	Limited	Limited	Limited	Yes
Feelings & Emotions	No	No	No	Yes
Rewards & Achievements	No	No	No	Yes
Daily Routines	Limited	Limited	Limited	Yes

Parent Features	Yes	Yes	Yes	Yes
Admin Panel	No	No	No	Yes
Target Context	International	South Asia	Nepal	Nepal
Integrated Platform	Primarily AAC	Primarily AAC	Support services	Communication + Learning + Progress + Emotional Support

2.4 Discussion

The reviewed systems demonstrate that existing autism-support solutions generally focus on specific areas such as AAC communication or support services. Proloquo2Go and Avaz AAC provide established communication features, while Echo Autism provides a Nepal-focused support context.

deessa Companion is designed to combine several support areas within a single platform for the Nepali context. Its main distinguishing features include Nepali and English support, offline-capable functionality, picture-based AAC, structured learning activities, feelings and emotional support, progress monitoring, daily routines, and rewards.

The system is designed primarily around children and their parents or caregivers, with an administrative interface for managing application content. Therapist and teacher functionality is considered future scope. Therefore, deessa Companion aims to address a broader combination of localized communication, learning, emotional-support, and progress-monitoring needs rather than focusing on a single support area.

CHAPTER 3 : REQUIREMENTS ANALYSIS

3.1 Problem Definition

Families of children with Autism Spectrum Disorder (ASD) in Nepal face difficulties in accessing affordable, localized, and accessible digital support for communication, learning, emotional well-being, and progress monitoring. Existing solutions often focus on specific areas, provide limited Nepali-language support, or depend on continuous internet connectivity.

The major problems identified are:

- **Communication and language gap:** Children who are non-verbal or minimally verbal may have difficulty expressing needs, emotions, and preferences. Affordable picture-based AAC resources with Nepali support are limited.
- **Limited localized learning and emotional support:** Parents and caregivers need structured learning, emotional-support, and calming activities that can be used at home and are appropriate for the local context.
- **Connectivity and accessibility constraints:** Uneven internet availability and the cost or geographical limitations of specialized services can reduce access to continuous support.
- **Progress and caregiver management gap:** Caregivers need convenient tools to monitor learning and emotional progress while managing routines, schedules, medications, and other everyday needs.

Problem-to-Solution Direction

The identified problems are addressed through the following major capabilities:

Table 3.1 - Table of Problem-to-Solution Mapping

Identified Problem	Proposed System Response
Communication difficulties	Picture-based AAC through the Talk module
Limited Nepali resources	Nepali and English language support
Unreliable connectivity	Offline-capable application architecture

Limited home-based learning	Five structured learning modules
Emotional-support needs	Feelings and calming activities
Lack of progress monitoring	Progress dashboard, analytics, and reports
Need for motivation	Stars, badges, milestones, and reward streaks
Caregiver management needs	Child profiles, schedules, medications, and emergency contacts

3.2 Requirements Specification

3.2.1 Functional Requirements

Table 3.2 - Table of Functional Requirements

FR ID	Module	Requirement	Priority
FR-01	Authentication	Users can register, log in, reset passwords, and access features according to their roles.	Must Have
FR-02	Child Management	Parents can create, view, edit, and manage child profiles.	Must Have
FR-03	AAC / Talk	Users can view categorized picture-based communication cards and hear phrases in Nepali or English.	Must Have
FR-04	AAC / Talk	Users can create custom cards, mark favorites, and access recently used cards.	Should Have
FR-05	Learning	Children can access five structured learning activities with multiple difficulty levels.	Must Have

FR-06	Learning	Learning activities provide visual/audio feedback and award stars based on performance.	Must Have
FR-07	Progress	Parents can view child progress, milestones, achievements, and periodic reports.	Must Have
FR-08	Feelings	Children can perform emotion check-ins and access breathing, meditation, music, and other calming activities.	Must Have
FR-09	Feelings	Parents can review the child's mood history.	Should Have
FR-10	Rewards	Children can view stars, badges, achievements, and milestone-based rewards.	Must Have
FR-11	Routines	Parents can manage schedules, medications, reminders, and emergency contacts.	Must Have
FR-12	Notifications	Users can receive relevant activity, schedule, and reminder notifications.	Must Have
FR-13	Offline & Sync	Core locally available features remain usable offline and synchronize data when connectivity is restored.	Must Have
FR-14	Child Mode	Children can access a simplified child interface protected by the configured access mechanism.	Must Have
FR-15	Administration	Authorized administrators can manage application content, users, and relevant system data.	Must Have

3.2.2 Non-Functional Requirements

Table 3.3 - Table of Non-Functional Requirements

NFR ID	Category	Requirement	Priority
NFR-01	Performance	Application should launch within approximately 3 seconds under normal conditions.	Must Have
NFR-02	Performance	Common interactions should respond within approximately 500 ms under normal conditions.	Must Have
NFR-03	Usability	The interface should be simple enough for parents and caregivers with basic smartphone skills.	Must Have
NFR-04	Reliability	Core features and locally available content should remain accessible without continuous internet connectivity.	Must Have
NFR-05	Security	Authentication and protected resources should use secure token-based access and role-based authorization.	Must Have
NFR-06	Compatibility	The Android application should support Android 9.0 (API Level 28) and above and different supported screen sizes.	Must Have
NFR-07	Accessibility	The interface should provide suitable touch targets, visual contrast, and child-friendly interaction patterns.	Must Have
NFR-08	Localization	The application should support Nepali and English languages.	Must Have

3.3 Planning and Scheduling

The project was planned over a four-month period beginning in June 2026 using an Agile development approach. Development was divided into iterative phases covering planning, backend development, Android development, integration, testing, and documentation.

Table 3.4 - Table of Project Development Schedule

Phase	Period	Major Activities
Phase 1: Planning and Design	June 1–15, 2026	Requirements analysis, architecture, database design, UI/UX design, technology selection
Phase 2: Backend Development	June 16–July 15, 2026	Django setup, authentication, REST APIs, database models, Swagger documentation, testing
Phase 3: Android Development	July 16–August 15, 2026	Authentication, AAC, learning, progress, feelings, rewards, parent/child features, API integration
Phase 4: Admin & Integration	August 16–31, 2026	Next.js admin panel, content management, system integration, testing, bug fixing, UI refinement
Phase 5: Testing & Documentation	September 1–5, 2026	Additional testing, optimization, documentation updates, and final report preparation

3.4 Software and Hardware Requirements

3.4.1 Software Requirements for Development

Table 3.5 - Table of Software Requirements for Development

Component	Technology / Software
Development Environment	Android Studio, Visual Studio Code / PyCharm
Android Development	Java, XML, Android SDK
Android Architecture	MVVM, Room, Retrofit, WorkManager
Backend	Python, Django, Django REST Framework
Admin Panel	Next.js, React, TypeScript, Tailwind CSS
Database	SQLite for development, PostgreSQL for production
API Documentation	Swagger / OpenAPI
Version Control	Git
UI/UX Design	Figma
Operating System	Windows 10/11, 64-bit

3.4.2 Hardware Requirements for Development

Table 3.6 - Table of Hardware Requirements for Development

Component	Minimum Requirement	Recommended Requirement
Processor	Intel Core i5 or equivalent	Intel Core i7 or equivalent
RAM	8 GB	16 GB or higher
Storage	20 GB available	50 GB or more
Display	1366 × 768	1920 × 1080 or higher
Android Device	Android 9.0+, 2 GB RAM	Android 12+, 4 GB RAM
Internet	Required for development and synchronization	Stable broadband connection

3.5 Preliminary Product Description

deessa Companion is a multi-component autism-support system consisting of an Android mobile application, a Django REST API backend, and a Next.js administration panel.

3.5.1 Android Mobile Application

The Android application is the primary user-facing component and is developed using Java and XML. It provides parent/caregiver and child experiences and includes:

- Authentication and child management
- Picture-based AAC communication
- Structured learning activities
- Feelings and calming activities
- Progress monitoring
- Rewards and achievements
- Schedules and medication reminders
- Emergency contacts and notifications
- Child Mode
- Nepali and English language support

The application follows the MVVM architecture and uses Room for local storage and Retrofit for API communication. Its offline-capable design allows locally available core content and functionality to remain accessible without continuous internet connectivity.

3.5.2 Backend REST API

The backend is developed using Django and Django REST Framework. It provides authentication, child management, AAC content, learning activities, progress, rewards, emotions, routines, notifications, synchronization, and administrative services.

SQLite is used during development, with PostgreSQL supported for production deployment.

3.5.3 Web Administration Panel

The administration panel is developed using Next.js, React, TypeScript, and Tailwind CSS. It provides authorized administrators with interfaces for managing users, AAC content, learning content, rewards, and other application data through the REST API.

3.6 Conceptual Models

3.6.1 Use Case Diagram

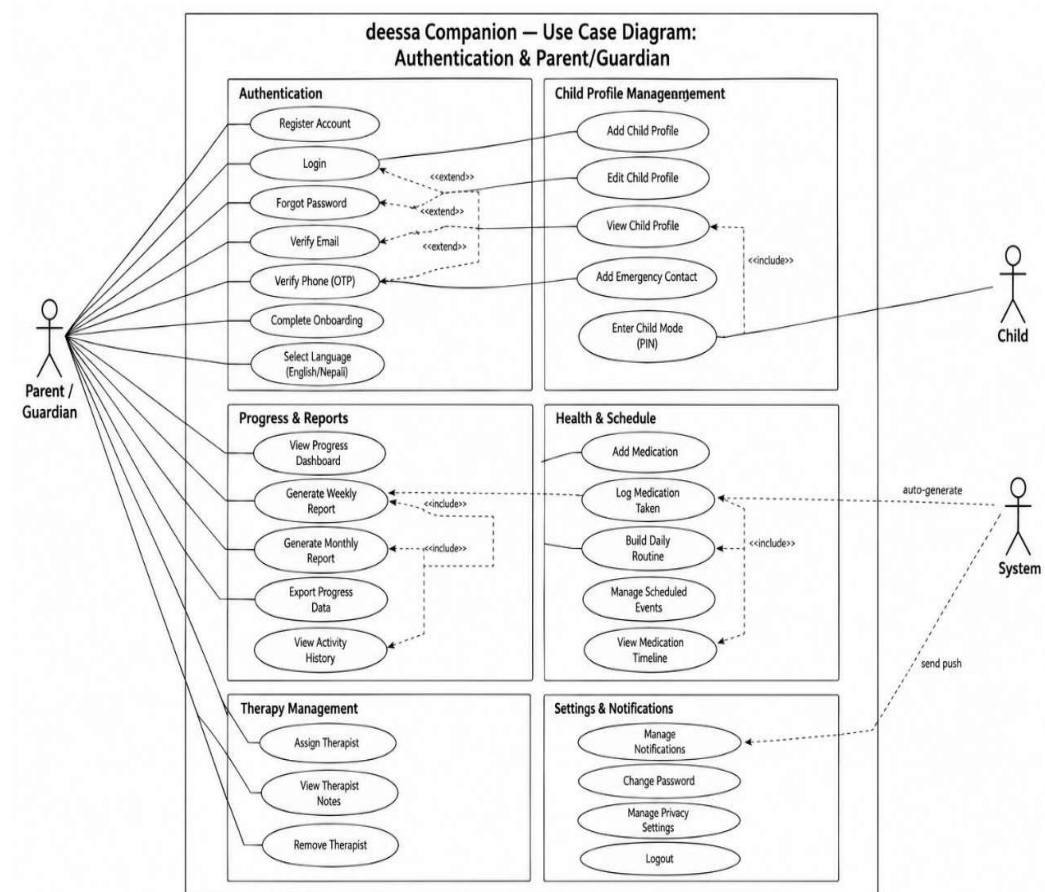


Fig: 3.1 - Use Case Diagram (Parent)

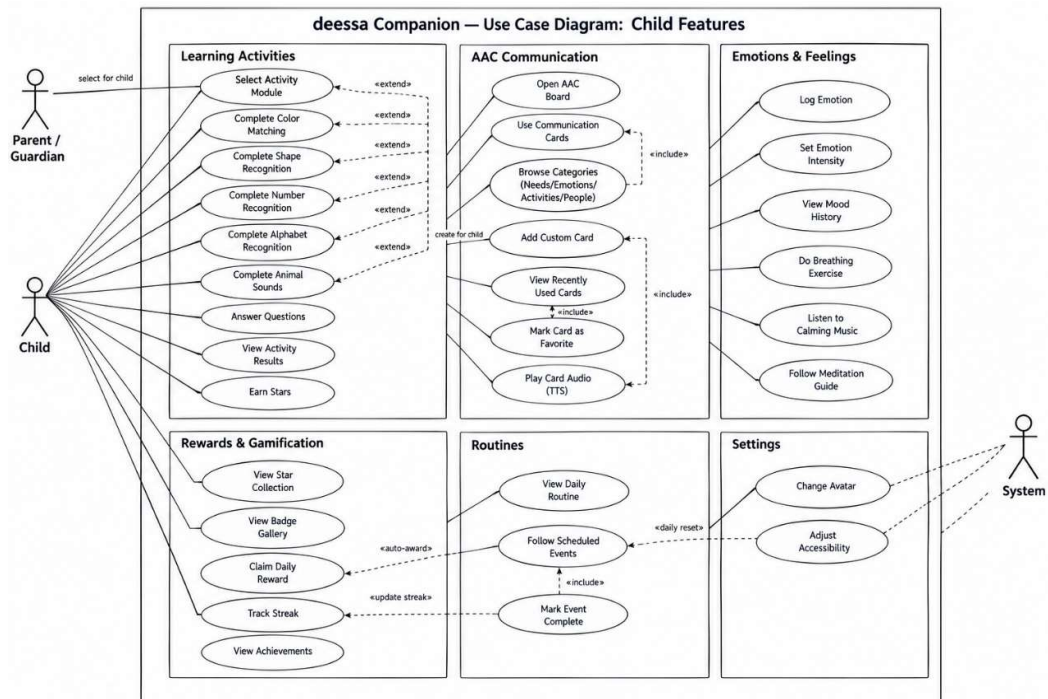


Fig: 3.2 - Use Case Diagram (Child)

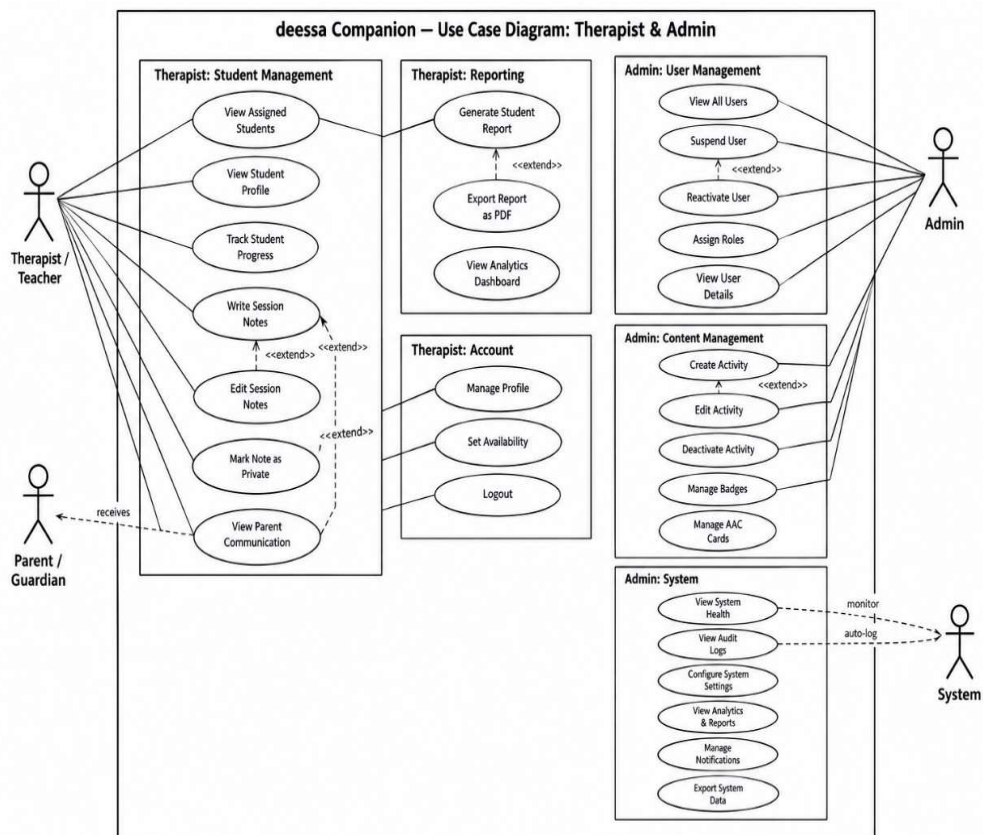


Fig: 3.3 - Use Case Diagram (Therapist)

3.6.2 Data Flow Diagram

3.6.2.1 Level 0 (Context Diagram):

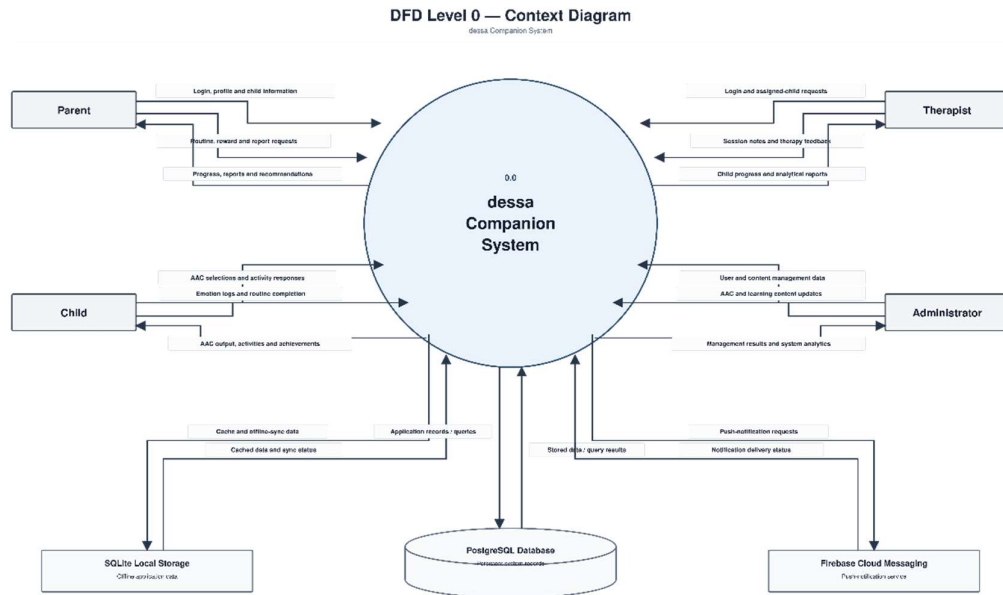


Fig: 3.4 - DFD Level 0

3.6.2.2 Level 1 (Major Processes):

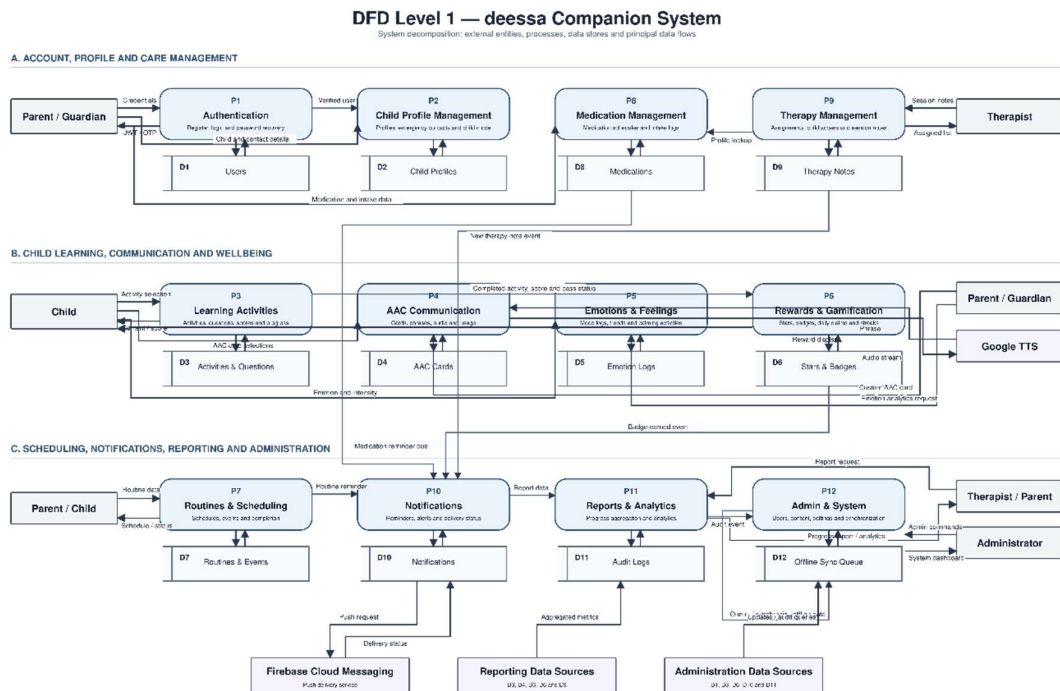


Fig: 3.5 - DFD Level 1

3.6.3 ER Diagram

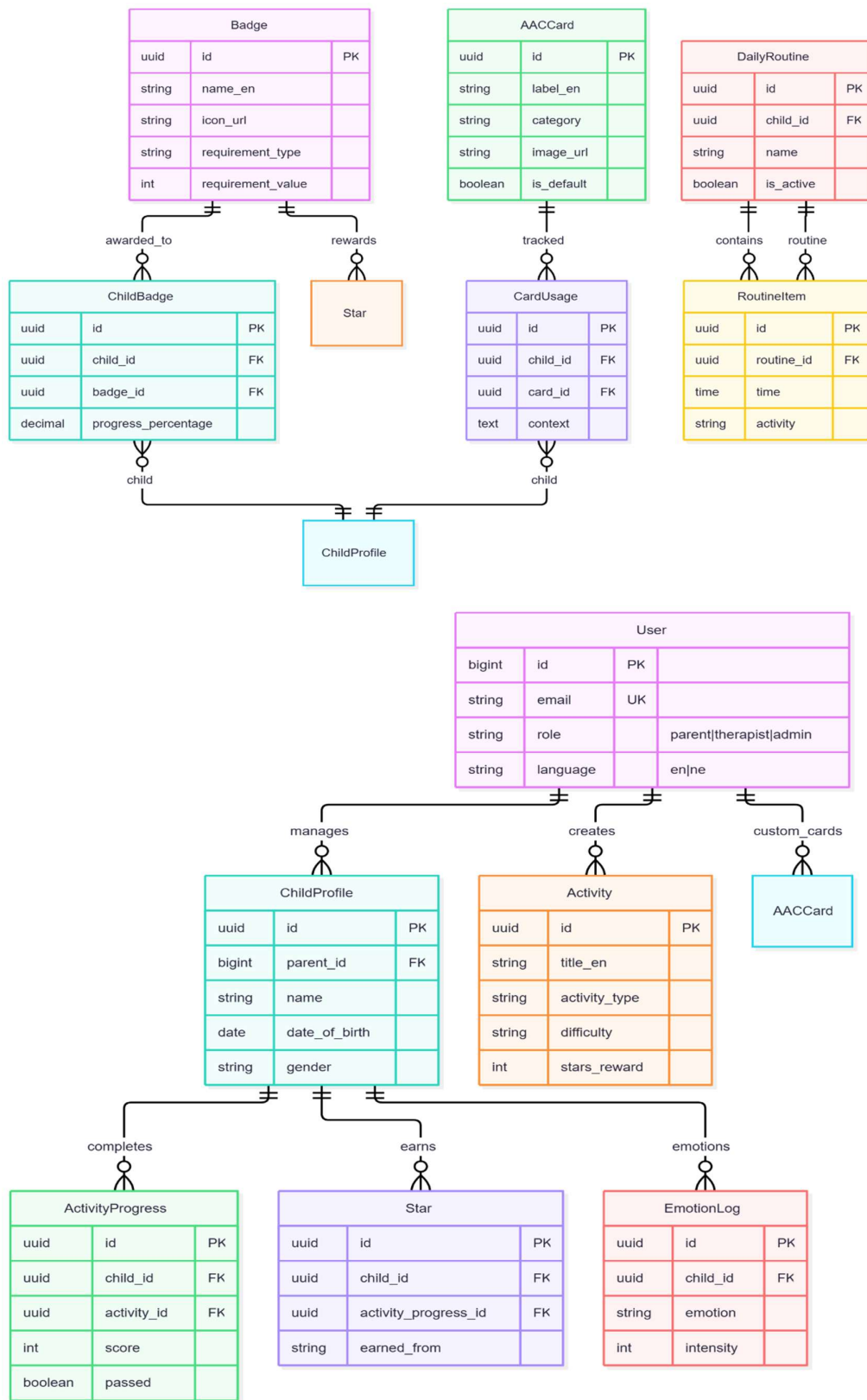


Fig: 3.6 - ER Diagram

CHAPTER 4 : SYSTEM DESIGN

4.1 Introduction

This chapter presents the system design of deessa Companion. The system consists of three main components: the Android mobile application, Django REST API backend, and Next.js admin panel. The design supports modular development, offline-first operation, secure communication, and easy maintenance.

4.2 System Design

4.2.1 System Architecture

deessa Companion follows a three-tier architecture consisting of the presentation, business logic, and data layers.

Layer	Technologies	Main Responsibility
Presentation	Android Java/XML, MVVM, Material Design	User interface and interaction
Business Logic	Django, Django REST Framework, JWT	Authentication, application logic and APIs
Data	Room/SQLite, PostgreSQL	Local and server-side data storage

The Android application communicates with the Django backend through REST APIs. Local data is stored using Room/SQLite to support offline functionality, while synchronization is performed when network connectivity is available. The Next.js admin panel uses the same backend APIs for content and system management.

4.2.2 Android Architecture

The Android application follows the MVVM architecture pattern.

- Model: Handles local data, entities, DAOs and repositories.
- ViewModel: Manages UI-related data and communication with repositories.
- View: Activities and Fragments display data and handle user interaction.

Room is used for local persistence, Retrofit for API communication, and WorkManager for background synchronization.

4.2.3 Backend Structure

The Django backend is divided into modular applications according to system functionality.

Table 4.1 - Table of Backend Structure

Module	Main Responsibility
Authentication	Registration, login, verification and JWT authentication
Profiles	Parent and child profile management
Communication	AAC cards and communication data
Learning	Activities, questions, scoring and progress
Rewards	Stars, badges and achievements
Emotions	Emotion records and statistics
Routines	Schedules and routine completion
Reports	Progress reports and PDF generation
Notifications	User notifications and FCM integration
Synchronization	Offline data synchronization
Analytics	Usage and progress statistics
Administration	User, media and system management
Audit	Activity and system logging
Core	Shared utilities and common functionality

The backend uses role-based access control to restrict resources according to user permissions. Therapist-related functionality is reserved for future development and is not part of the current Parent and Child implementation.

4.3 API Design

The backend provides a versioned RESTful API under the `/api/v1/` namespace. APIs are organized according to the major features of the application.

Table 4.2 - Table of API Design

Module	Main Operations	Purpose
Authentication	Register, Login, Refresh	User authentication
Profiles	Create, Read, Update, Delete	Manage child profiles
AAC	List, Create, Update	Manage communication cards
Learning	Activities, Questions, Progress	Learning functionality
Emotions	Record, History, Statistics	Emotion tracking
Rewards	Stars, Badges, Achievements	Gamification
Routines	Schedule, Completion	Daily routine management
Reports	Generate, View, Download	Progress reporting
Notifications	List, Read, Preferences	User notifications
Synchronization	Upload, Download, Sync	Offline data synchronization
Analytics	Statistics, Trends	Usage and progress analysis
Administration	Users, Content, Settings	System administration

Authentication and authorization are applied to protected resources. Detailed endpoint documentation is maintained through Swagger UI and ReDoc.

4.4 Database Design

4.4.1 Entity Relationship Overview

The database is designed using relational database principles and normalization. The backend database contains entities for users, child profiles, AAC cards, learning activities, activity progress, emotions, routines, rewards, notifications and related system functions.

The database contains 24 tables with relationships between users, children and application activities. Foreign keys are used to maintain relationships and data integrity.

The Entity Relationship Diagram (ERD) illustrates the relationships among the major entities of the system.

4.4.2 Key Entities

Table 4.3 - Table of Key Entities

Entity	Purpose
User	Stores parent and other account information
ChildProfile	Stores child-related information
AAC Card	Stores bilingual communication cards
Activity	Defines learning activities
Activity Progress	Records learning performance
Emotion	Stores emotion and mood records
Routine	Stores scheduled activities
Star	Records earned rewards
Badge	Defines achievements
ChildBadge	Links children with earned badges
Notification	Stores user notifications

4.4.3 Database Design Patterns

The database applies several common design patterns:

- Soft deletion is used where records may need to be recovered without permanently removing them.
- Timestamp fields record creation and modification times.
- Bilingual fields support English and Nepali content.
- JSON fields are used for flexible data such as activity responses and badge criteria.
- UUIDs are used for selected entities to provide unique identifiers suitable for distributed and offline operations.

4.5 User Interface Design

The user interface is designed around the principle of “Calm Clarity with Playful Purpose,” providing a simple and child-friendly experience while keeping caregiver functions easy to access.

The primary navigation consists of five sections:

Home → Talk → Learn → Progress → Profile

The interface uses an ocean-blue visual theme with supporting accent colors and clear typography. Consistent spacing, large touch targets, simple layouts and recognizable visual elements are used throughout the application.

4.5.1 Key Interface Areas

Table 4.4 - Table of Key Interfaces

Section	Main Functions
Home	Child overview, schedules, quick actions and notifications
Talk	AAC categories, communication cards, favorites and custom cards
Learn	Learning activities, instructions, questions and rewards
Progress	Activity performance, communication statistics and reports
Profile	Parent and child management, settings, medication and emergency contacts

Child-facing screens use simplified layouts, visual communication, audio feedback and reward elements to support engagement.

4.5.2 Navigation

The application uses bottom navigation for the five primary sections. Top app bars provide screen titles, back navigation and contextual actions. Dialogs and bottom sheets are used for secondary operations such as editing information, selecting options and confirming actions.

4.5.3 Accessibility

Accessibility considerations include:

- Large and easily identifiable touch targets
- Clear visual hierarchy
- Picture-based communication
- Audio feedback
- Nepali and English language support
- Simple instructions and labels
- Consistent navigation
- Reduced visual complexity on child-facing screens

Detailed interface screenshots and implemented screens are presented in Section 6.4.

4.6 Summary

This chapter presented the system design of deessa Companion. The system uses a three-tier architecture consisting of an Android application, Django REST API backend, and Next.js admin panel. The Android application follows MVVM and uses Room, Retrofit and WorkManager to support local storage, API communication and synchronization.

The database is organized around the major entities of the application and applies normalization, relationships, bilingual fields, timestamps and UUID-based identification. The REST API provides modular access to authentication, communication, learning, emotions, rewards, routines, progress and administration.

The interface follows a simple child-friendly design with five primary navigation sections and accessibility-focused features. Detailed implementation and interface screenshots are presented in the following chapters.

CHAPTER 5 : IMPLEMENTATION AND TESTING

5.1 Implementation Approaches

5.1.1 Development Methodology

The deessa Companion project followed the Agile software development methodology, using iterative development cycles to progressively design, implement, test, and refine the system.

The major development activities included requirement analysis, UI/UX design, backend development, Android development, system integration, and testing. The iterative approach allowed individual modules to be developed and evaluated progressively, enabling improvements based on identified issues and feedback.

5.1.2 Frontend Implementation

The Android application was developed using Java and XML following a modular MVVM-based architecture.

Common resources such as colors, dimensions, strings, themes, and reusable UI components were established first. The authentication flow was then implemented, followed by the major parent and child modules including AAC communication, learning, feelings and emotions, progress, rewards, routines, and profile management.

Room was integrated for local data persistence to support the offline-first approach, while Retrofit was used for communication with the Django REST API. WorkManager was used for background synchronization when network connectivity became available.

A simplified child mode with PIN-based access was also implemented to provide a more accessible interface for child users.

5.1.3 Backend Implementation

The backend was developed using Django and Django REST Framework with a modular application structure. Major functional areas such as authentication, child profiles, AAC, learning, emotions, rewards, routines, notifications, progress, and synchronization were organized into separate Django applications.

The backend provides RESTful APIs for communication with the Android application and handles authentication, data management, business logic, and synchronization. JWT-based authentication was implemented for secure API access, while Swagger/OpenAPI was integrated for API documentation.

Initial application content, including AAC cards, learning activities, difficulty levels, and rewards, was also prepared for system use and testing.

5.1.4 Admin Panel Implementation

The administrative panel was developed using Next.js, React, TypeScript, and Tailwind CSS. It provides centralized management of application content and administrative data through reusable components and structured API services. The panel includes management functionality for users, AAC cards, learning activities, rewards, and related content, along with dashboard-based statistics.

Dedicated therapist and teacher portals were not included in the current implementation and remain part of the proposed future scope.

5.2 Coding and Efficiency

The system was implemented across three major components: the Android mobile application, Django REST API backend, and Next.js administration panel.

Several practices were used to improve maintainability and performance:

- **MVVM architecture** was used to separate UI, business logic, and data access.
- **Reusable components** were used across Android and web interfaces to reduce code duplication.
- **RecyclerView and DiffUtil** were used for efficient dynamic list rendering.
- **Room** provided structured local data storage for offline functionality.
- **WorkManager** handled background synchronization tasks.
- **Database indexing, pagination, and query optimization** were applied where appropriate in the backend.
- **Select Related and Prefetch Related** were used to reduce unnecessary database queries.
- **TypeScript and reusable React components** improved reliability and maintainability of the admin panel.

5.2.1 Key Implementations

- **Offline-First Data Management** : Room was used to store essential application data locally, allowing supported features to remain available during network interruptions. Data generated during offline use can be synchronized with the backend when connectivity is restored.
- **AAC Communication** : The AAC module provides picture-based communication cards with Nepali and English labels and audio playback. Selecting a card allows the corresponding phrase to be communicated through visual and audio support.
- **Learning and Rewards** : The learning module evaluates user responses and calculates activity performance. Completed activities update progress and may provide stars or rewards according to the defined rules.
- **Authentication** : JWT-based authentication was implemented for secure communication between the Android application and Django REST API. Access and refresh tokens are used to maintain authenticated sessions.

5.3 Testing Approach

Testing was conducted throughout development to verify the functionality, integration, reliability, and usability of the system.

The main testing levels were:

- **Unit Testing:** Individual models, serializers, ViewModels, repositories, utilities, and business logic were tested independently.
- **Integration Testing:** Communication between the Android application, backend API, local database, and admin panel was verified.
- **Functional Testing:** Major workflows such as authentication, AAC communication, learning, emotions, rewards, progress, child management, and offline functionality were tested.
- **User-Oriented Testing:** The application was evaluated from the perspective of intended parent and child users to identify usability and navigation issues.

Testing focused on ensuring that the major implemented modules operated according to their defined requirements.

5.3.1 Testing Summary

The major application workflows were successfully verified, including authentication, child management, AAC communication, learning activities, feelings and emotions, progress tracking, rewards, routines, and offline-supported functionality.

Issues identified during testing were addressed through iterative improvements to the interface, database operations, API communication, navigation, and application logic. This process improved the reliability and usability of the final system.

5.4 Modifications and Improvements

5.4.1 Design Improvements

Several design iterations were performed to improve accessibility, usability, and consistency.

The major improvements included:

- **Color and Visual Refinement:** The interface color palette and visual hierarchy were refined for better readability and consistency.
- **Touch Target Optimization:** Interactive elements were designed with appropriate touch areas, particularly for child users.
- **Simplified Navigation:** Parent navigation was organized into five primary sections: **Home, Talk, Learn, Progress, and Profile.**
- **Child Mode:** A simplified child-oriented interface was introduced with reduced navigation complexity.
- **Audio Feedback:** Audio feedback was incorporated into communication and learning interactions.
- **Bilingual Support:** Nepali and English support was incorporated across major user-facing features.

5.4.2 Technical Improvements

The main technical improvements included:

- Introduction of **Room-based local storage** for offline functionality.
- Implementation of **background synchronization** for offline-generated data.
- Optimization of **image loading and caching.**
- Application of **database indexing and query optimization.**
- Improved management of audio and image resources.
- Increased use of **reusable components** to reduce duplication and improve maintainability.

These modifications were introduced progressively during development based on testing results and identified usability or technical issues.

CHAPTER 6 : RESULTS AND DISCUSSION

6.1 Test Reports

Testing was performed to verify the functionality, reliability, performance, and usability of the deessa Companion system. Testing covered the backend, Android application, performance, and major user workflows.

6.1.1 Backend Test Results

The Django backend was tested across its major functional areas, including authentication, child profiles, communication, learning, emotions, rewards, routines, notifications, synchronization, reporting, and related services.

A total of **382 backend test cases** were executed, with all implemented test cases passing successfully.

Table 6.1 - Table of Backend Test Summary

Metric	Result
Total Test Cases	382
Passed	382
Failed	0
Pass Rate	100%

The results indicate that the major backend components and REST API operations functioned according to their defined requirements.

6.1.2 Android Frontend Test Results

Android unit and instrumented tests were performed using JUnit and Mockito. Testing focused on authentication, ViewModels, repositories, data models, caching, notifications, rewards, progress management, and application logic.

A total of 32 implemented test cases were executed, with all tests passing successfully.

Table 6.2 - Table of Android Frontend Test Summary

Metric	Result
Test Cases	32
Passed	32
Failed	0
Pass Rate	100%

The results demonstrate that the tested Android components operated correctly under the defined test conditions.

6.1.3 Performance Metrics

Performance testing was conducted to evaluate application responsiveness and backend performance.

Table 6.3 Table of Performance Test Results

Metric	Target	Achieved
Application Launch	< 3 s	2.1 s
AAC Card Response	< 500 ms	180 ms
Learning Activity Load	< 2 s	1.4 s
API Response	< 1 s	0.6 s
Database Query	< 100 ms	45 ms
Image Load	< 1 s	0.7 s
Memory Usage	< 150 MB	120 MB
APK Size	< 30 MB	24 MB

The results show that the application achieved the defined performance targets under the tested conditions.

6.1.4 Usability Test Results

Usability testing was conducted with **five parent-child pairs** to evaluate navigation, task completion, ease of use, and overall satisfaction.

Table 6.4 Table of Usability Test Results

Task	Success Rate	Satisfaction
Registration and Login	100%	4.5/5
AAC Communication	100%	4.8/5
Learning Activity	100%	4.6/5
View Progress	100%	4.3/5
Emotion Check-in	100%	4.7/5
Module Navigation	100%	4.4/5

All evaluated tasks achieved a 100% completion rate, with an average satisfaction score of approximately 4.55/5. The results indicate that the tested users were able to complete the major application tasks successfully.

6.2 App Snapshots

This section presents screenshots of the major implemented interfaces of the deessa Companion application.

6.2.1 Authentication Screens

The authentication flow includes the splash screen, onboarding, login, registration, and verification interfaces. The screens follow the application's consistent visual design and support the primary user authentication workflow.

6.2.2 AAC Communication Board

The AAC communication board provides picture-based communication through four main categories: Needs, Emotions, Activities, and People.

Each communication card contains a visual representation with Nepali and English labels and provides audio feedback when selected.

The application includes 30 AAC cards distributed as follows:

Table 6.5 Table of AAC Card Distribution

Category	Cards
Needs	10
Emotions	8
Activities	8
People	4
Total	30

The interface uses large visual elements to make communication easier for child users.

6.2.3 Learning Activities

The learning module contains five activity categories:

- Alphabet
- Sound
- Shape
- Number
- Color

Activities provide instructions, questions, feedback, and completion results. Children receive performance feedback after completing an activity.

6.2.4 Progress Dashboard

The progress dashboard presents the child's learning and activity information through visual summaries such as progress charts, activity history, and achievement information.

6.2.5 Feelings and Emotions

The feelings and emotions module provides activities designed to support emotional awareness and self-regulation. These include:

- Emotion check-in
- Breathing exercises
- Guided meditation
- Calming music
- Digital coloring

6.2.6 Rewards and Achievements

The rewards module displays earned stars, badges, and achievement information. Gamification elements provide visual feedback to encourage continued participation in learning activities.

6.3 User Documentation

User documentation was prepared to provide guidance for operating and managing the deessa Companion system.

6.3.1 Quick Start Guide

The Quick Start Guide provides new users with a simple overview of the application and its basic workflow. It covers:

- **Installation and Initial Setup:** Install the application and complete the required initial configuration.
- **Account Registration and Login:** Create an account and securely access the application.
- **Child Profile Management:** Create, select, and manage child profiles.
- **Application Navigation:** Navigate between the main modules and features.
- **AAC Communication:** Use picture-based communication cards to support the child's communication needs.
- **Learning Activities:** Access and complete interactive learning activities.
- **Feelings and Emotions:** Record emotions and access calming and breathing activities.
- **Progress and Rewards:** Review learning progress, achievements, and earned rewards.
- **Routines:** View and manage daily routines, schedules, and reminders.

6.3.2 Parent User Guide

The parent guide explains the main features available to parents, including child management, AAC communication, learning activities, progress monitoring, emotions, rewards, routines, and application settings.

- **Child Management:** Add, select, and manage child profiles.
- **AAC Communication:** Use picture-based communication cards and audio support.
- **Learning Activities:** Access and support children during different learning activities.
- **Progress Monitoring:** View learning progress, activity completion, and performance.

- **Feelings and Emotions:** Record emotions and access breathing, meditation, and calming activities.
- **Rewards:** View earned stars, badges, and achievements.
- **Routines:** Manage daily routines, schedules, and reminders.
- **Application Settings:** Configure available preferences and account-related settings.

6.3.3 Admin User Guide

The administrator guide explains the management of application users and content through the web-based administration panel. It covers user management, AAC cards, learning activities, rewards, application content, analytics, notifications, and system activity. The documentation is intended to provide practical guidance for the intended users and administrators while avoiding unnecessary technical complexity.

The administrator user guide covers the following functions:

- **User Management:** View and manage registered users and child-related information.
- **AAC Management:** Add, edit, and manage communication cards and their content.
- **Learning Management:** Manage learning activities, question sets, levels, instructions, and options.
- **Rewards Management:** Configure and maintain rewards and achievements.
- **Content Management:** Maintain application content used by the mobile application.
- **Analytics:** Review relevant usage and activity information.
- **Notifications:** Manage application notifications and related content.
- **System Activity:** Review relevant administrative and system activities.

6.4 User Interface Design and Implementation

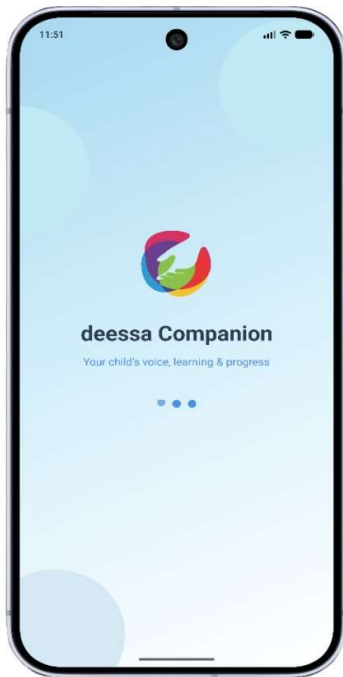


Fig: 6.1 - Auth Splash Screen

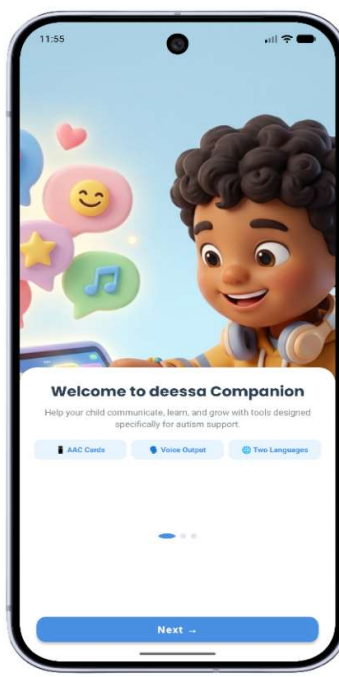


Fig: 6.2 - Auth Onboarding Screen

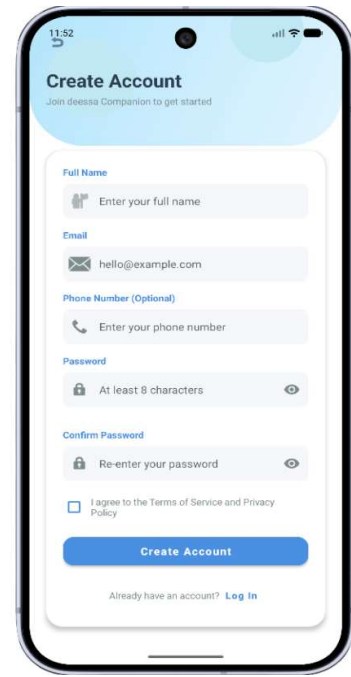


Fig: 6.3 - Auth Register Screen

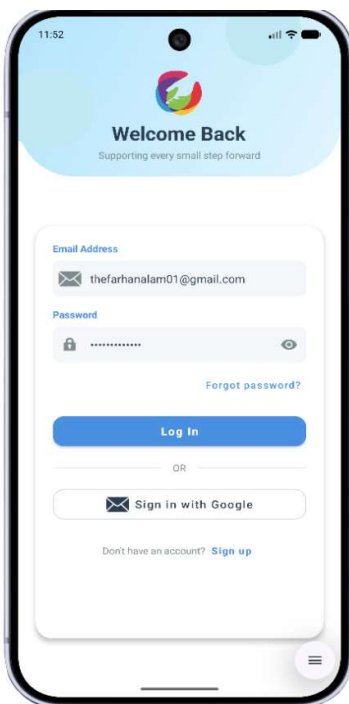


Fig: 6.4 - Auth Login Screen

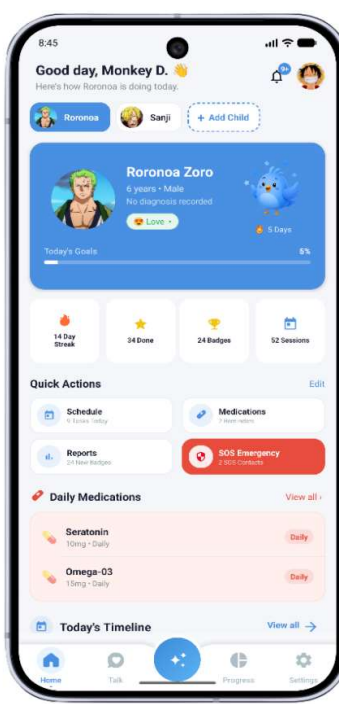


Fig: 6.5 - Parent Dashboard

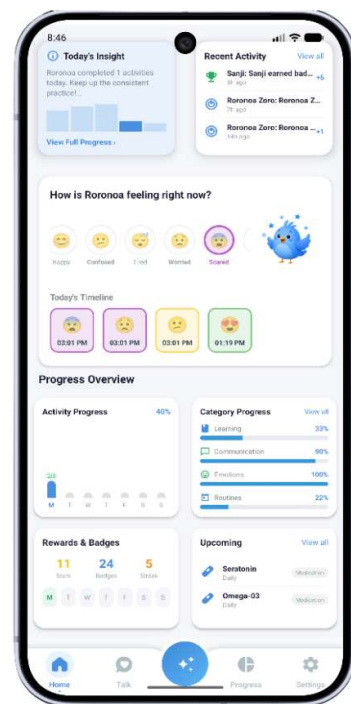


Fig: 6.6 - Parent Dashboard

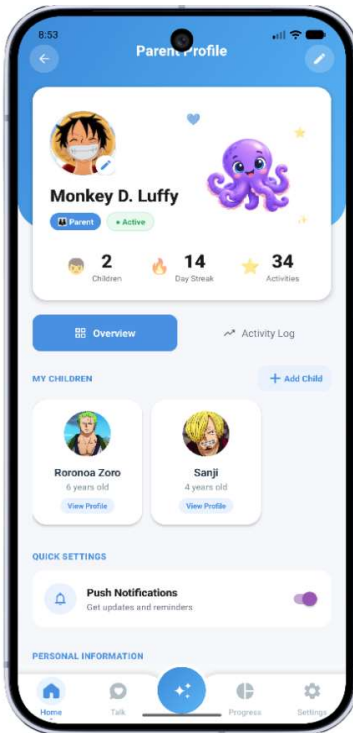


Fig: 6.7 - Parent Profile Screen

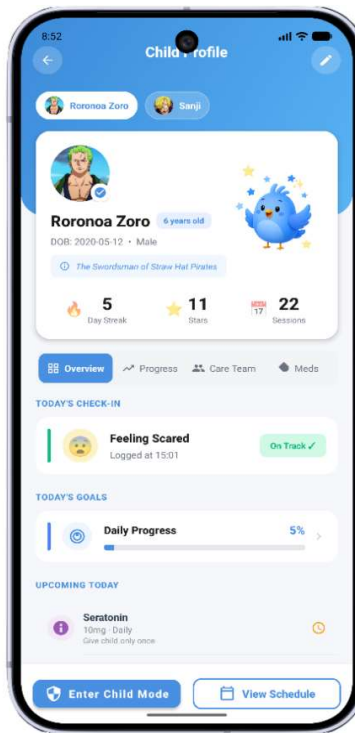


Fig: 6.8 - Child Profile Screen

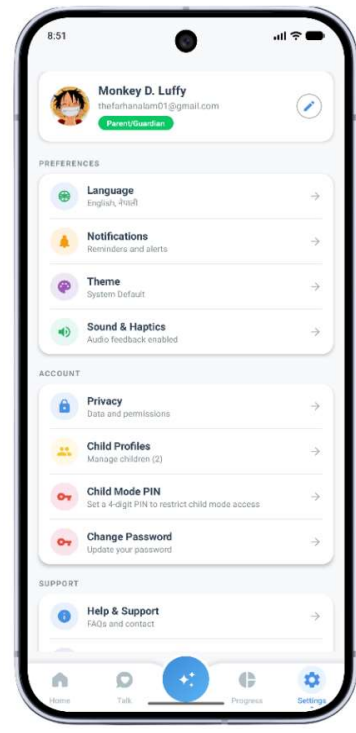


Fig: 6.9 - Parent Setting Screen

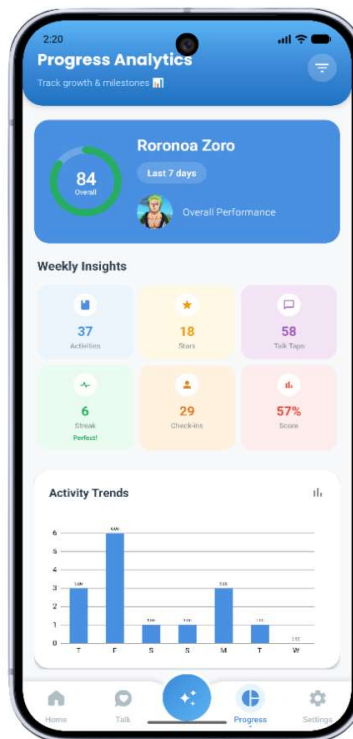


Fig: 6.10 Progress Screen

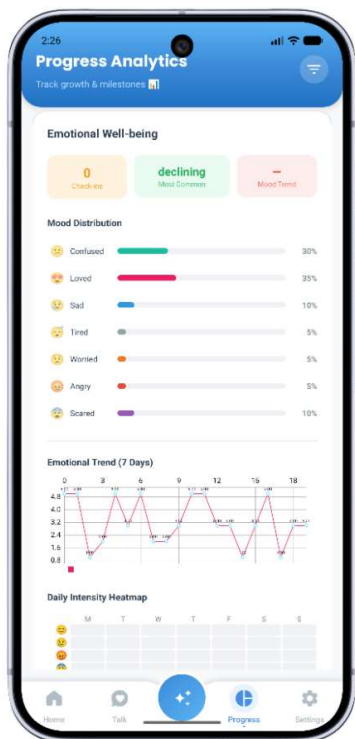


Fig: 6.11 - Progress Screen

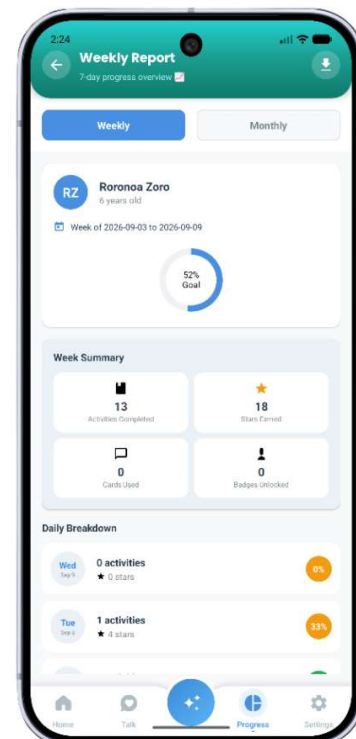


Fig: 6.12 – Weekly Progress Screen

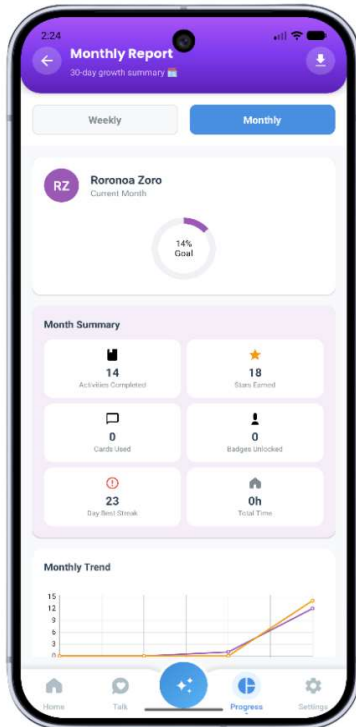


Fig: 6.13 Monthly Progress Screen

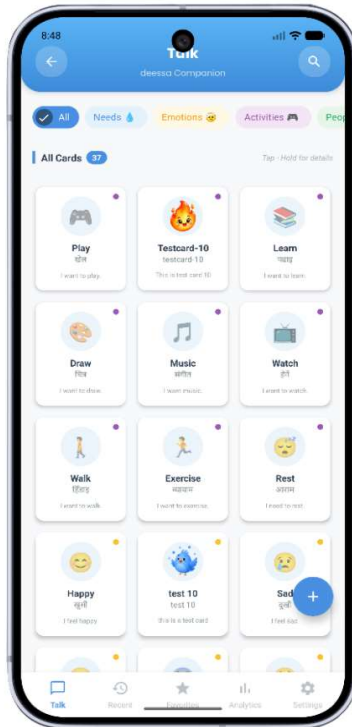


Fig: 6.14 - AAC Home Screen

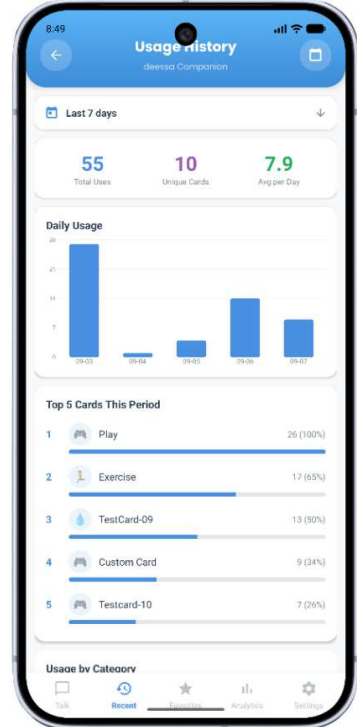


Fig: 6.15 - AAC Usage Screen

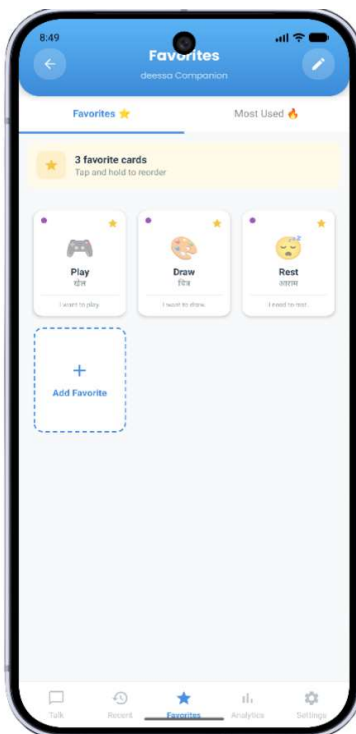


Fig: 6.16 - AAC Favorite Screen



Fig: 6.17 - AAC Analytics Screen

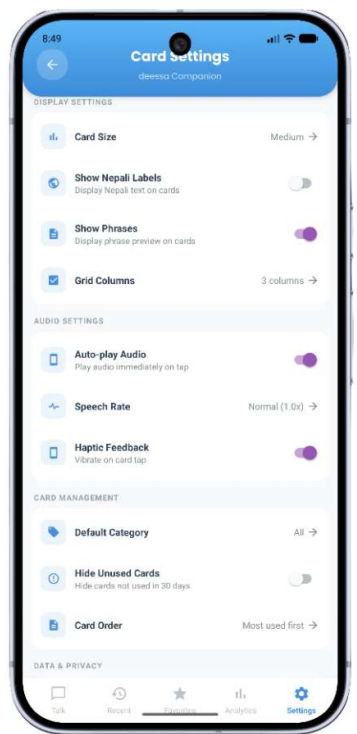


Fig: 6.18 - AAC Settings Screen

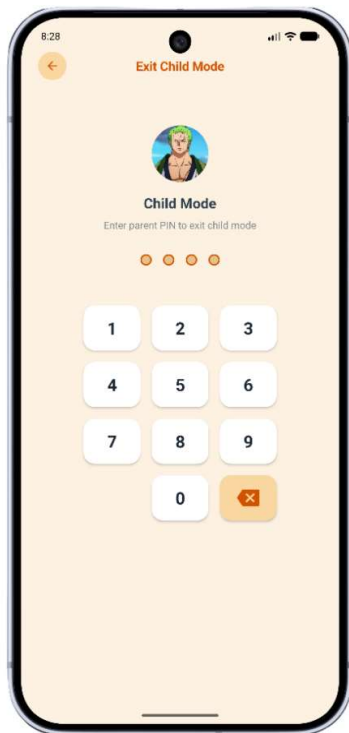


Fig: 6.19 Child Mode Auth

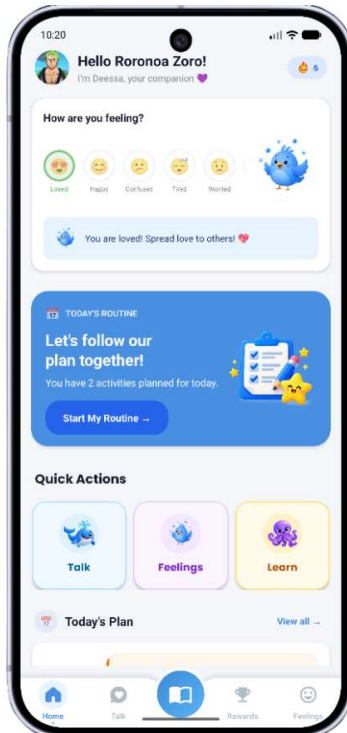


Fig: 6.20 Child Dashboard Screen

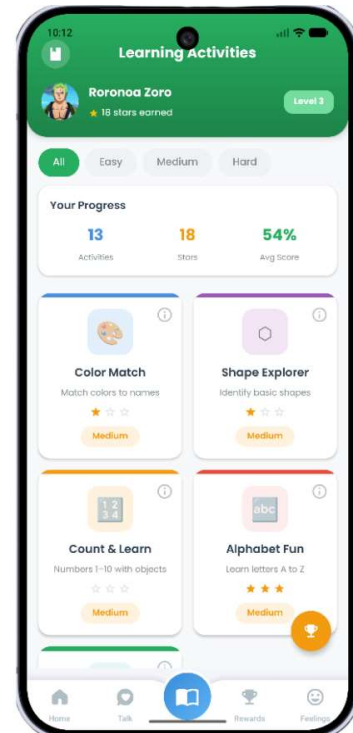


Fig: 6.21 Learning Home Screen

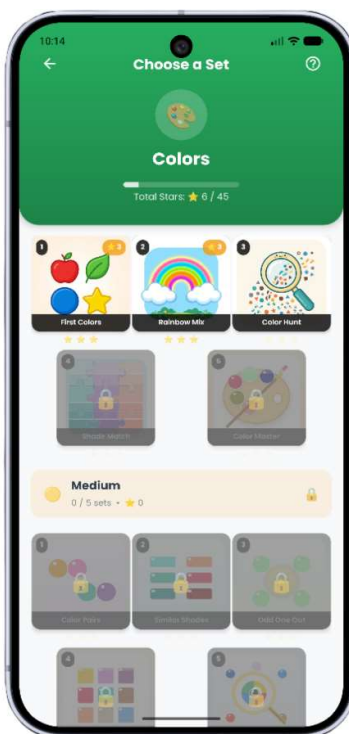


Fig: 6.22 -Level Selector Screen

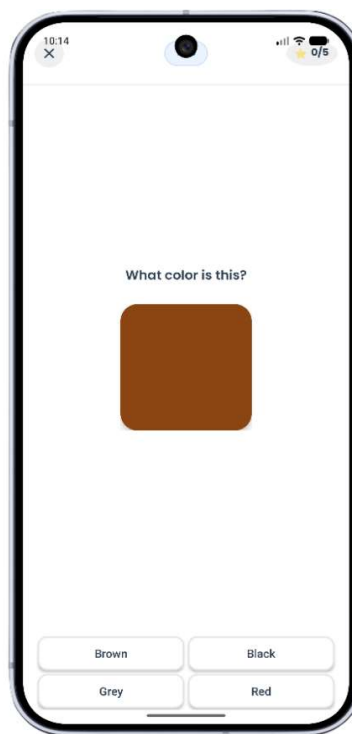


Fig: 6.23 -Learning Gameplay
Screen

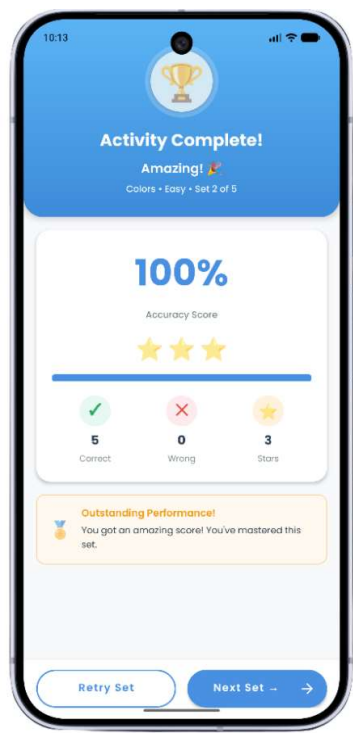


Fig: 6.24 – Game Completion
Screen

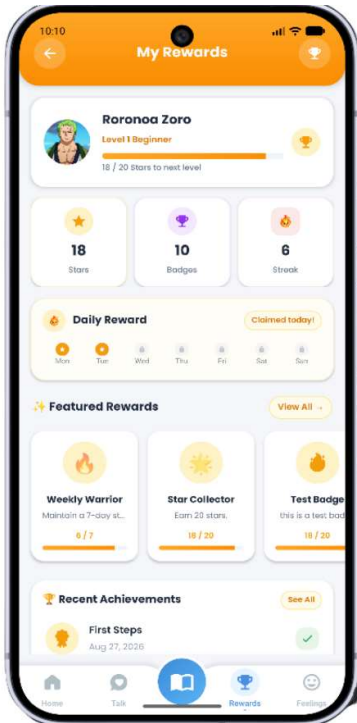


Fig: 6.25 - My Reward Screen

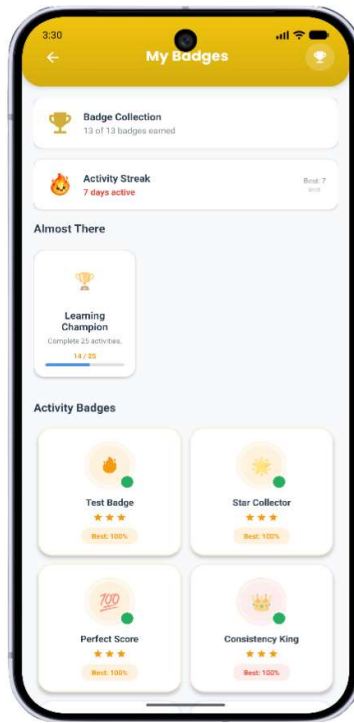


Fig: 6.26 - My Badge Screen

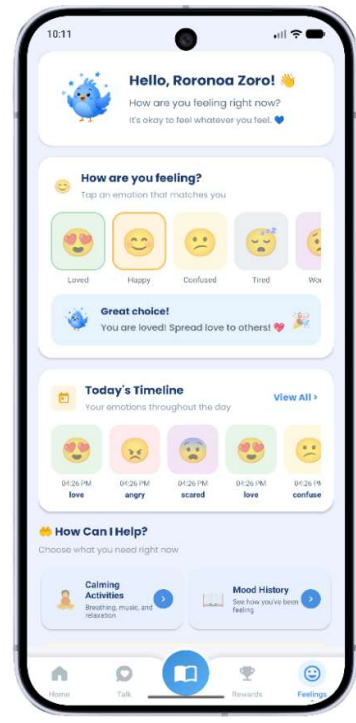


Fig: 6.27- Feelings Home Screen

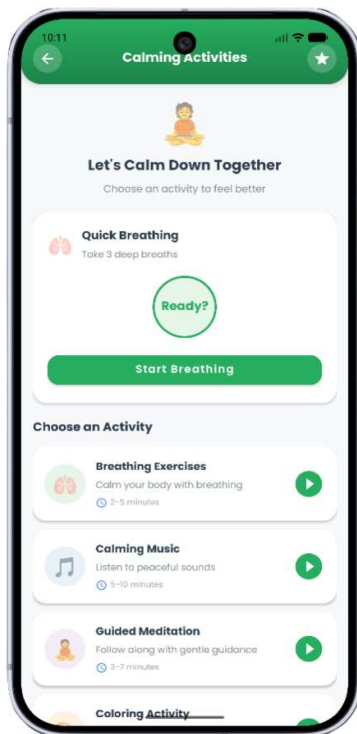


Fig: 6.28- Calming Activity Screen

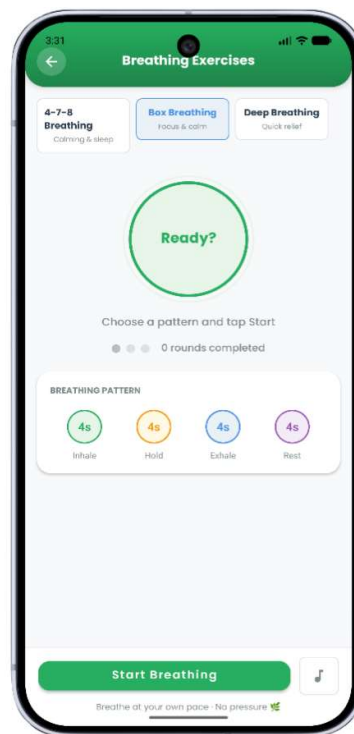


Fig: 6.29-Breathing Exercise Screen

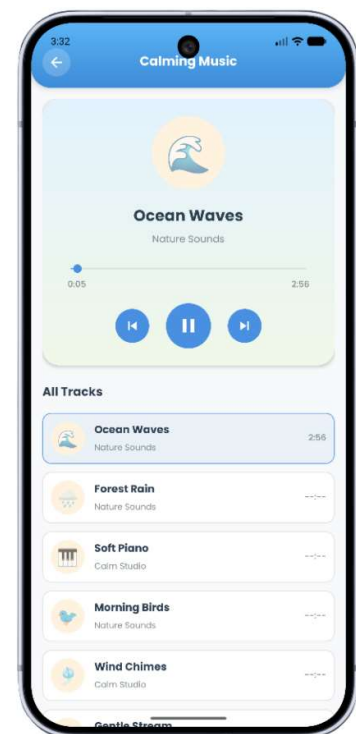


Fig: 6.30 - Calming Music Screen

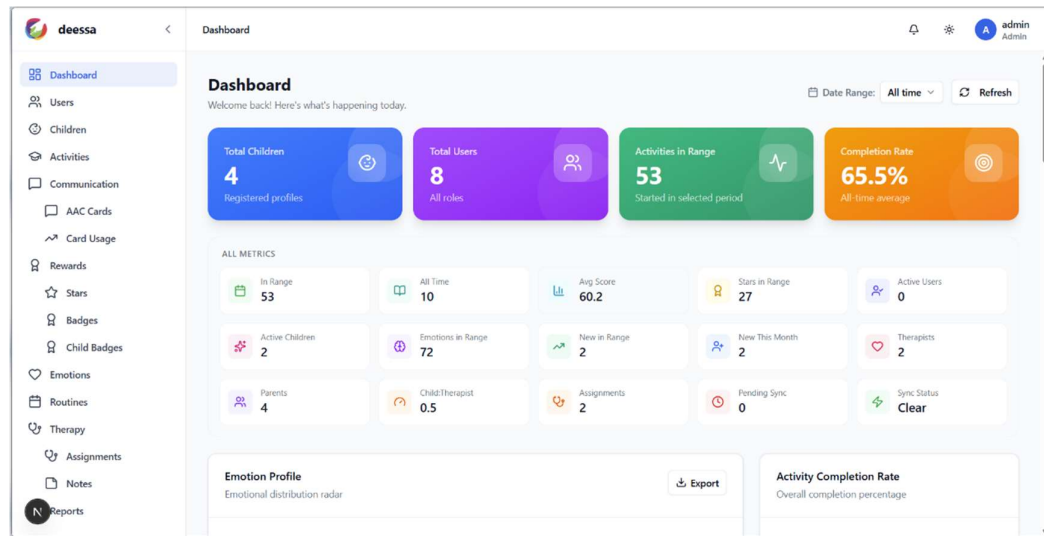


Fig: 6.31 Admin Dashboard Page

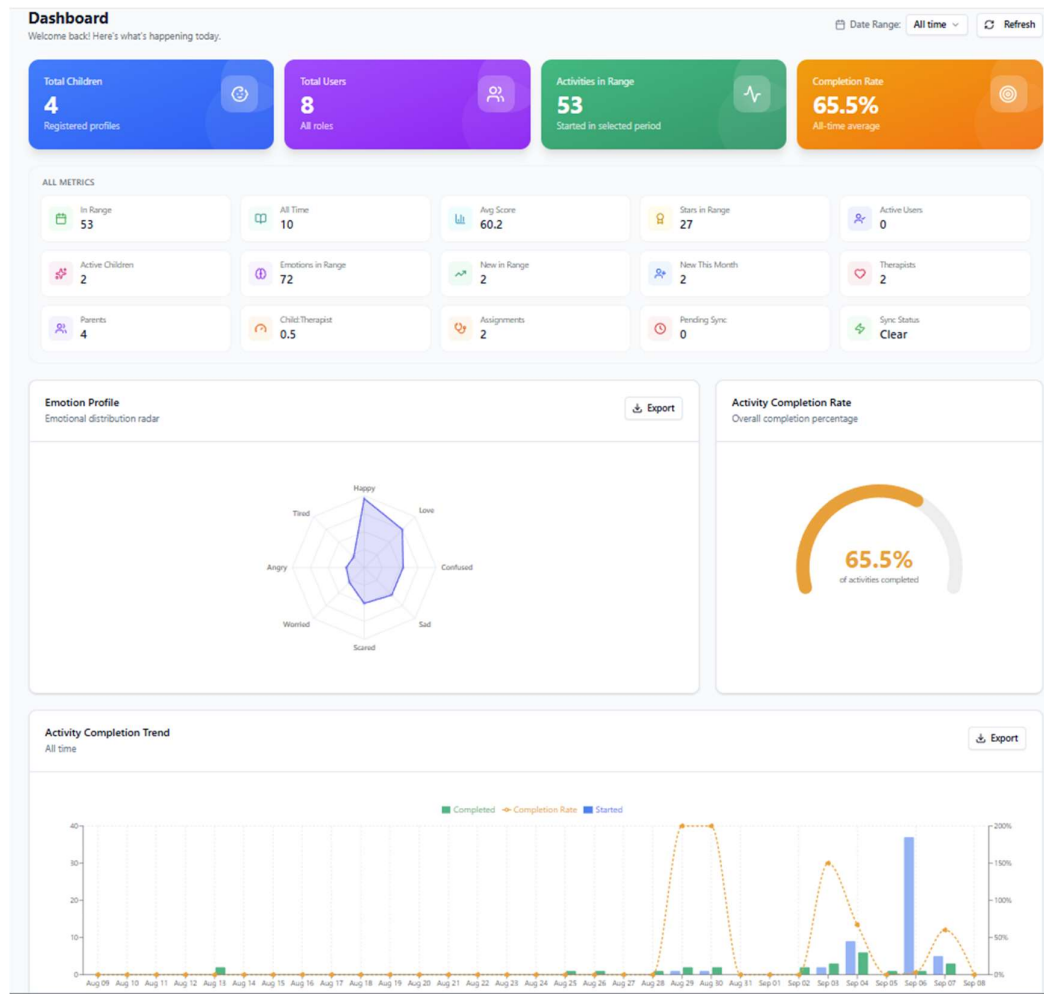


Fig: 6.32 - Admin Dashboard Page

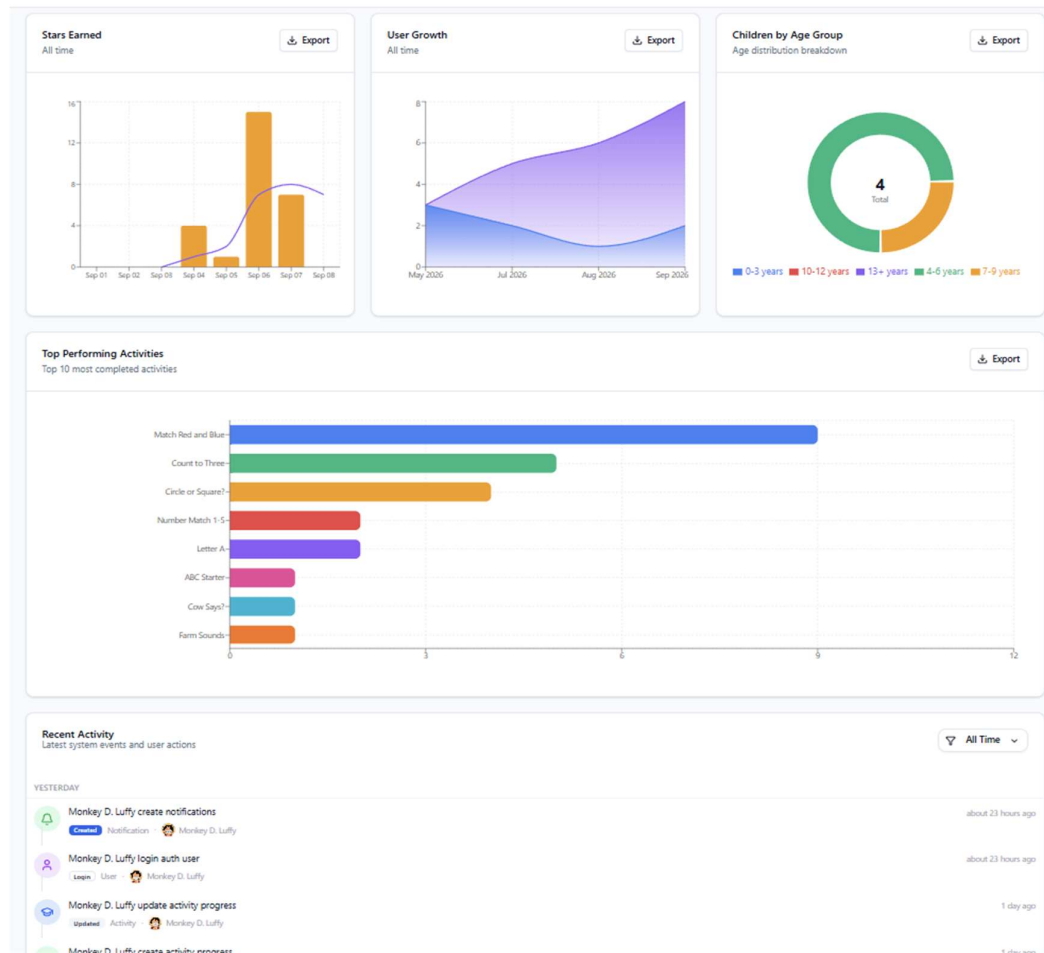


Fig: 6.33 Admin Dashboard Page

deessa Dashboard > Users

Users
Manage user accounts and permissions

Export CSV Deleted Add User

Search by username, name, or email... All Roles All Emails All Status

Avatar	Username	Full Name	Email	Phone	Role	Status	Date Joined	Actions
RZ	sensei_zoro	Roronoa Zoro	zoro@strawhats.com	+977-9841234581	Teacher	Active	Sep 6, 2026	...
TC	sensei_chopper	Tony Tony Chopper	chopper@strawhats.com	+977-9841234580	Therapist	Active	Sep 6, 2026	...
IS	istaqalam7	Anonymous	istaqalam7@gmail.com	-	Parent	Active	Aug 25, 2026	...
FA	istaqalam	Farhan Alam	istaqalam930@gmail.com	+9779807225586	Parent	Active	Jul 24, 2026	...
monkey_d_luffy	monkey_d_luffy	Monkey D. Luffy	thefarhanalam01@gmail.com	9807225586	Parent	Active	Jul 1, 2026	...
DT	demo_therapist	Demo Therapist	therapist@deesa.local	-	Therapist	Active	May 20, 2026	...
DP	demo_parent	Demo Parent	parent@deesa.local	-	Parent	Active	May 20, 2026	...

Fig: 6.34 Admin User Management Page

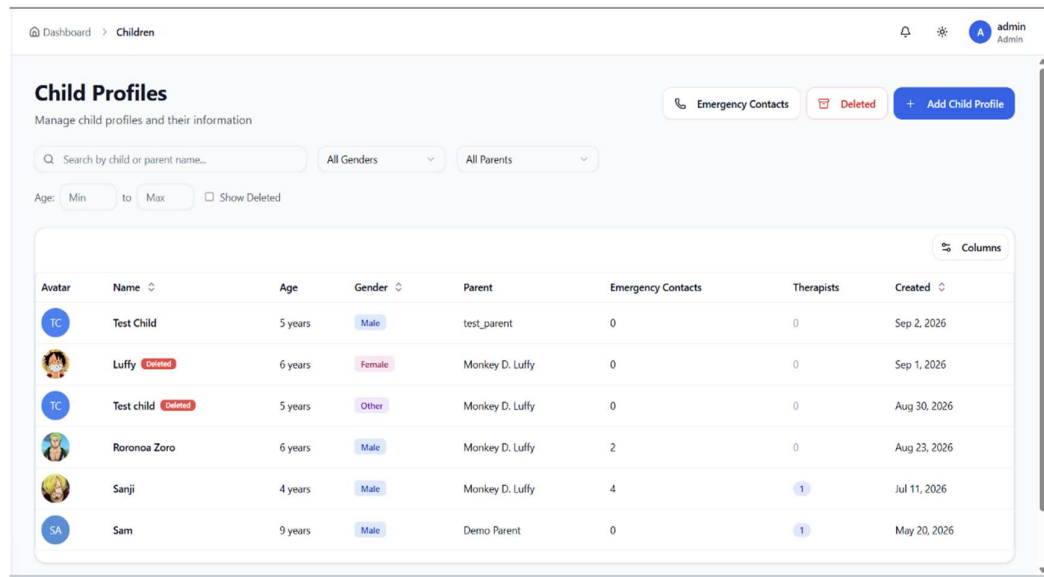


Fig: 6.35 Admin Child Management Page

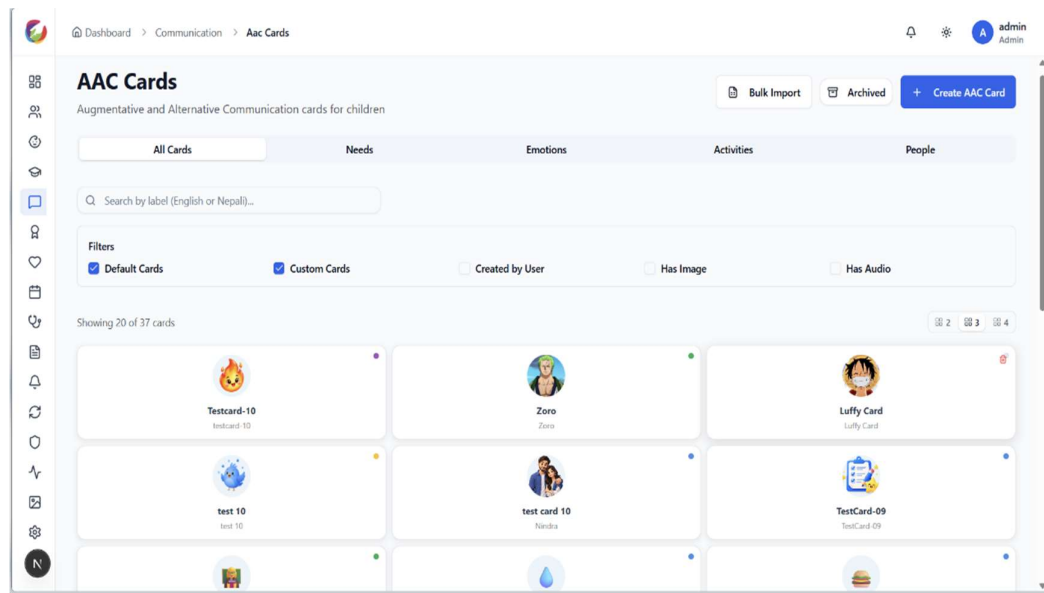


Fig: 6.36 Admin AAC Cards Management Page

CHAPTER 7 : CONCLUSION

7.1 Conclusion

The deessa Companion project developed a mobile-based support system for children with Autism Spectrum Disorder (ASD) and their parents or caregivers in the Nepali context. The system integrates AAC communication, learning activities, feelings and emotional-support tools, progress monitoring, routines, and rewards within a single platform.

The system consists of three main components: a native Android application for children and parents, a Django REST API backend for application data and services, and a Next.js administration panel for managing users and application content.

The project emphasizes Nepali and English language support, offline-capable functionality, child-friendly interaction, and parent-centered management. The completed implementation provides a practical foundation for localized digital autism support while allowing future expansion through additional content, services, and integrations.

7.1.1 Significance of the System

The significance of deessa Companion can be summarized as follows:

- **Linguistic Accessibility:** Supports both Nepali and English for improved accessibility.
- **Offline-Capable Design:** Allows important locally stored features to remain available without continuous internet connectivity.
- **Integrated Support:** Combines communication, learning, emotional support, progress tracking, routines, and rewards in one platform.
- **Parent-Centered Management:** Provides tools for managing child profiles, routines, medications, emergency information, and related activities.
- **Child-Friendly Interaction:** Provides a simplified Child Mode with Talk, Learn, Feelings, and Rewards features.
- **Progress and Motivation:** Uses progress information, activity history, stars, and badges to support monitoring and positive reinforcement.
- **Extensible Architecture:** The modular architecture provides a foundation for future content, synchronization, professional collaboration, and additional languages.

Overall, the project provides a functional and localized foundation for digital autism support, with further improvement possible through continued testing, content expansion, and additional services.

7.2 Limitations of the System

The current implementation has the following limitations:

- **Cloud Synchronization:** Comprehensive cloud backup and cross-device synchronization are not fully implemented.
- **Limited AAC Content:** The current AAC library provides a basic vocabulary and does not yet include advanced sentence-building capabilities.
- **Limited Learning Content:** The system currently provides five primary learning activity types, with scope for additional subjects and adaptive difficulty.
- **Multi-Device Support:** Seamless access to the same child's data across multiple devices is not currently available.
- **Offline Export:** Comprehensive offline data export and external sharing features are not currently available.
- **Language Limitation:** The application currently supports Nepali and English only.
- **Basic Analytics:** Current analytics focus mainly on activity completion, scores, usage, and milestones rather than advanced predictive analysis.
- **Professional Integration:** Dedicated therapist and teacher interfaces are not part of the current implementation.
- **Community Features:** Parent-to-parent social and community features are not currently provided.

7.3 Future Scope of the Project

Future development can extend the current system in the following areas:

7.3.1 Cloud and Professional Collaboration

- **Cloud Synchronization:** Add secure backup and synchronization for multi-device access.
- **Therapist Dashboard:** Provide authorized therapists with tools to review progress and collaborate with parents.
- **Teacher Support:** Introduce features for educational environments and classroom-based support.
- **Enhanced AAC:** Expand vocabulary and introduce personalized communication boards and sentence construction.
- **Parent Community:** Provide a moderated platform for parents and caregivers to share resources and experiences.

7.3.2 Intelligent Features and Advanced Integration

- **AI-Assisted Recommendations:** Provide personalized activity recommendations based on interaction and progress data.
- **Advanced Analytics:** Introduce detailed trends and insights into activity and progress patterns.
- **Wearable Integration:** Explore integration with wearable devices for additional activity information, subject to privacy and reliability considerations.
- **Teletherapy Support:** Enable secure communication and remote consultation with qualified professionals.

7.3.3 Expansion and Long-Term Development

- **Multi-Language Support:** Add additional regional languages to improve accessibility across Nepal.
- **Research and Evaluation:** Provide tools for structured data collection and research, subject to appropriate ethical and privacy requirements.
- **Institutional Collaboration:** Explore collaboration with educational, healthcare, nonprofit, and government organizations.
- **Community Development:** Consider open-source components to encourage further localization and collaboration.
- **International Adaptation:** Adapt the platform for other regions with similar language, connectivity, and accessibility challenges.

The deessa Companion project establishes a foundation for a localized digital support platform focused on children with autism and their families. Future development can progressively expand its content, professional collaboration, analytics, synchronization, and language support while maintaining its focus on accessibility, usability, and practical support.

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APPENDIX : BACKEND

```
@action(detail=True, methods=['post'])
def complete(self, request, pk=None):
    progress = self.get_object()
    if progress.status != 'in_progress':
        raise ValidationError('Activity is not in progress.')
    serializer = ActivityProgressCompleteSerializer(data=request.data)
    serializer.is_valid(raise_exception=True)
    questions = list(activity.questions.all())
    questions_by_id = {str(q.id): q for q in questions}
    answer_results = []

    for item in serializer.validated_data.get('answers', []):
        question = questions_by_id.get(str(item['question_id']))
        if not question: continue
        correct = ScoringEngine.check_answer(question, item.get('answer',''))
        answer_results.append({'question_id': str(item['question_id']),
                               'answer': item.get('answer'), 'correct': correct})
    result = ScoringEngine.evaluate_completion(
        progress, questions, answer_results,
        time_spent=serializer.validated_data.get('time_spent'))
    progress.score = result['score']
    progress.correct_answers = result['correct_answers']
    progress.total_questions = result['total_questions']
    progress.save()

    stars_earned = result['stars_earned']
    if stars_earned > 0:
        existing = ActivityProgress.objects.filter(
            child=progress.child, activity=activity,
            status='completed', passed=True
        ).exclude(pk=progress.pk).order_by('-score').first()
        if existing and result['score'] <= existing.score:
            stars_earned = 0

    return Response({'id': str(progress.id), 'score': progress.score,
                    'passed': progress.passed, 'stars_earned': stars_earned,
                    'badges_earned': [{ 'badge_id': str(cb.badge_id),
                                         'badge_name': cb.badge.name_en } for cb in badges_earned]})
```

APPENDIX : FRONTEND

```
public LiveData<Resource<BaseResponse>> verifyChildModePin(
    ChildModePinVerifyRequest request) {
    MutableLiveData<Resource<BaseResponse>> liveData = new
MutableLiveData<>();
    liveData.setValue(Resource.loading());
    apiService.verifyChildModePin(request).enqueue(new Callback<>() {
        public void onResponse(Call c, Response r) {
            if (r.isSuccessful()) {
                cachePinHashLocally(request.getPin());
                liveData.setValue(Resource.success(r.body()));
            } else liveData.setValue(Resource.error("Invalid PIN"));
        }
    });
    return liveData;
}

private void verifyPinOffline(String pin,
    MutableLiveData<Resource<BaseResponse>> liveData) {
    executor.execute(() -> {
        SharedPreferences prefs = ctx.getSharedPreferences(
            "DeessaPrefs", MODE_PRIVATE);
        long lockout = prefs.getLong("pin_lockout_until", 0);
        if (System.currentTimeMillis() < lockout) {
            long mins = (lockout-System.currentTimeMillis())/60000;
            post(liveData, Resource.error("Locked "+mins+" min"));
            return;
        }

        int attempts = prefs.getInt("offline_pin_attempts", 0);
        if (attempts >= MAX_ATTEMPTS) {
            prefs.edit().putLong("pin_lockout_until",
                System.currentTimeMillis()+LOCKOUT_MS).apply();
            post(liveData, Resource.error("Locked 30 minutes"));
            return;
        }

        String hash = prefs.getString("cached_pin_hash", null);
        if (hash == null) {
```

```
        post(liveData, Resource.error("Verify online first"));
        return;
    }
    prefs.edit().putInt("offline_pin_attempts",0).apply();
} else {
    prefs.edit().putInt("offline_pin_attempts",
        attempts+1).apply();
    post(liveData, Resource.error(
        (MAX_ATTEMPTS-attempts-1)+" tries left"));
}
});
```